Theory of Mathematics

Volume II: Real Analysis and Topological Foundations

Contents

Part I

Metric Foundations

Chapter 1

The Real Number System and Completeness

The real numbers are an ordered field completed by a least-upper-bound principle. Completeness converts approximation into existence and drives the convergence arguments used throughout analysis.

Learning goals

The reader should be able to use the chapter's definitions in proofs, recognize the decisive hypotheses, and move between concrete metric examples and invariant topological statements.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{structural theorem} \longrightarrow \text{application.}}$$

1.1 Ordered fields

Order is compatible with addition and multiplication. Positivity, trichotomy, and cancellation permit algebraic manipulation of inequalities.

1.2 Bounds and extrema

Upper and lower bounds describe sets globally; maxima and minima are bounds attained by elements of the set.

1.3 Suprema and infima

The supremum is the least upper bound and the infimum the greatest lower bound. Neither need belong to the set.

1.4 Completeness

Every nonempty real set bounded above has a supremum. This property fails in the rational numbers.

1.5 Archimedean property

No real number dominates all positive integers; equivalently, reciprocals of positive integers can be made arbitrarily small.

1.6 Density

Between distinct real numbers lie both rational and irrational numbers.

1.7 Absolute value and inequalities

Absolute value is the one-dimensional norm and satisfies triangle and reverse-triangle inequalities.

1.8 Nested intervals and roots

Completeness yields nested-interval principles and existence of positive square roots.

1.9 Core structural results

Theorem 1.1 (Archimedean property)**.** *For every real x there is $n \in \mathbb{N}$ with $n > x$.*

Proof. Otherwise $\mathbb{N}$ would have a supremum s; then $s - 1$ would fail to be an upper bound, producing $n > s - 1$ and hence $n + 1 > s$, contradiction. ☐

Theorem 1.2 (Supremum approximation)**.** *If $s = \sup A$ and $\varepsilon > 0$, some $a \in A$ satisfies $s - \varepsilon < a \leq s$.*

Proof. If not, $s - \varepsilon$ would itself be an upper bound, contradicting the leastness of s. ☐

Theorem 1.3 (Square-root existence)**.** *For every $a > 0$ there is a unique $r > 0$ with $r^2 = a$.*

Proof. Take $r = \sup\{x \geq 0 : x^2 \leq a\}$. If $r^2 < a$, a small increase remains admissible; if $r^2 > a$, a small decrease remains an upper bound. Both contradict the definition of r. ☐

1.10 Worked examples

Example 1.4 (Supremum of a quadratic set)**.** Let $A = \{x \geq 0 : x^2 < 5\}$. Identify $\sup A$. The set is nonempty and bounded. Its supremum s must satisfy $s^2 = 5$ by the square-root supremum argument, so $s = \sqrt{5}$.

Example 1.5 (Infimum by reflection)**.** Prove $\inf A = -\sup(-A)$ for nonempty A bounded below. Lower bounds of A correspond under sign change to upper bounds of $-A$; reversing the order turns the least upper bound into the greatest lower bound.

Example 1.6 (No largest rational below an irrational)**.** If α is irrational, show $\{q \in \mathbb{Q} : q < \alpha\}$ has no maximum. Given $q < \alpha$, density of $\mathbb{Q}$ gives a rational r with $q < r < \alpha$.

Example 1.7 (Reciprocals become small)**.** Given $\varepsilon > 0$, prove some n satisfies $1/n < \varepsilon$. Choose $n > 1/\varepsilon$ by the Archimedean property and take reciprocals.

1.11 Solved dossiers

Each dossier is a substantial solved problem. Corpus mappings guide topic choice where explicit retained source blocks exist; otherwise the dossier is a fresh canonical completion.

Problem 1.8 (Supremum of a quadratic set)**.** Let $A = \{x \geq 0 : x^2 < 5\}$. Identify $\sup A$.

Solution. The set is nonempty and bounded. Its supremum s must satisfy $s^2 = 5$ by the square-root supremum argument, so $s = \sqrt{5}$. ☐

Problem 1.9 (Infimum by reflection)**.** Prove $\inf A = -\sup(-A)$ for nonempty A bounded below.

Solution. Lower bounds of A correspond under sign change to upper bounds of $-A$; reversing the order turns the least upper bound into the greatest lower bound. ☐

Problem 1.10 (No largest rational below an irrational)**.** If α is irrational, show $\{q \in \mathbb{Q} : q < \alpha\}$ has no maximum.

Solution. Given $q < \alpha$, density of $\mathbb{Q}$ gives a rational r with $q < r < \alpha$. ☐

Problem 1.11 (Reciprocals become small)**.** Given $\varepsilon > 0$, prove some n satisfies $1/n < \varepsilon$.

Solution. Choose $n > 1/\varepsilon$ by the Archimedean property and take reciprocals. ☐

Problem 1.12 (Integer bracketing)**.** Show every $x \in \mathbb{R}$ satisfies $m \leq x < m + 1$ for some $m \in \mathbb{Z}$.

Solution. Use the Archimedean property to restrict to finitely many candidate integers and choose the largest integer not exceeding x. ☐

Problem 1.13 (Rational density)**.** Prove that $x < y$ implies there is rational q with $x < q < y$.

Solution. Choose n with $1/n < y - x$ and then an integer m with $nx < m \leq nx + 1$; m/n lies between x and y. ☐

Problem 1.14 (Irrational density)**.** Prove there is an irrational number between any two reals.

Solution. Choose rational r in $(x - \sqrt{2}, y - \sqrt{2})$ and use $r + \sqrt{2}$. ☐

Problem 1.15 (Reverse triangle inequality)**.** Prove $||x| - |y|| \leq |x - y|$.

Solution. Apply the triangle inequality to $x = (x - y) + y$ and then interchange x, y. ☐

Problem 1.16 (Nested intervals are unique when lengths vanish)**.** If nested closed intervals have lengths tending to zero, show their intersection has at most one point.

Solution. Two common points would have distance bounded by every interval length, hence by arbitrarily small positive numbers, so they coincide. ☐

Problem 1.17 (Why $\mathbb{Q}$ is incomplete)**.** Explain why $\{q \in \mathbb{Q} : q^2 < 2, q \geq 0\}$ has no rational supremum.

Solution. Its real supremum is $\sqrt{2}$, and any rational candidate below or above can be improved using the order argument; $\sqrt{2} \notin \mathbb{Q}$. ☐

Problem 1.18 (Geometric partial sums)**.** Find the supremum of $1 + 1/2 + \cdots + 1/2^n$.

Solution. The sum equals $2 - 2^{-n}$, so 2 is an upper bound approached arbitrarily closely and hence is the supremum. ☐

Problem 1.19 (Completeness synthesis)**.** Explain how a supremum packages arbitrarily accurate lower approximations into one number.

Solution. Completeness supplies $s = \sup A$, and supremum approximation places elements of A within every positive distance below s. ☐

1.12 Exercises with complete solutions

The shorter exercise layer is kept separate from the dossiers.

Exercise 1.20. Find $\sup\{n/(n+1) : n \geq 1\}$.

Hint. Compare with 1. ☐

Solution. The supremum is 1. ☐

Exercise 1.21. Find $\inf\{1/n : n \geq 1\}$.

Hint. Use Archimedean smallness. ☐

Solution. The infimum is 0. ☐

Exercise 1.22. Solve $|2x - 1| \leq 3$.

Hint. Use a double inequality. ☐

Solution. The solution is $[-1, 2]$. ☐

Exercise 1.23. Show $|x + y| \geq ||x| - |y||$.

Hint. Apply reverse triangle inequality. ☐

Solution. It is the reverse triangle inequality with x and $-y$. ☐

Exercise 1.24. Give a bounded rational set with no rational supremum.

Hint. Use $q^2 < 3$. ☐

Solution. $\{q \in \mathbb{Q} : q^2 < 3, q \geq 0\}$ works. ☐

Exercise 1.25. Show every finite nonempty real set has a maximum.

Hint. Induct on the cardinality. ☐

Solution. Adjoin one element at a time and take the larger of it and the previous maximum. ☐

Exercise 1.26. If $A \subset B$, compare suprema.

Hint. Every upper bound of B bounds A. ☐

Solution. When both exist, $\sup A \leq \sup B$. ☐

Exercise 1.27. If $|x - y| < \varepsilon$ for every $\varepsilon > 0$, show $x = y$.

Hint. Assume positive distance. ☐

Solution. Take $\varepsilon = |x - y|/2$ to obtain a contradiction. ☐

Chapter summary

The central definitions, estimates, and structural theorems of this chapter will be used repeatedly in the remaining analysis volume.

Chapter 2

Euclidean and Normed Spaces

Normed spaces add quantitative geometry to vector spaces. The norm axioms isolate exactly the estimates needed to speak about distance, convergence, boundedness, and continuity.

Learning goals

The reader should be able to use the chapter's definitions in proofs, recognize the decisive hypotheses, and move between concrete metric examples and invariant topological statements.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{structural theorem} \longrightarrow \text{application.}}$$

2.1 Euclidean norm

The Euclidean norm is induced by the standard inner product and measures ordinary length.

2.2 Norm axioms

A norm is positive definite, absolutely homogeneous, and subadditive.

2.3 Standard finite-dimensional norms

The 1-, 2-, and infinity norms emphasize total magnitude, Euclidean length, and largest coordinate.

2.4 Balls and spheres

Norm balls define neighborhoods and therefore the topology of a normed space.

2.5 Operator norms

The induced norm of a linear map measures its maximal stretching on the unit sphere.

2.6 Equivalent norms

Comparable norms define the same convergence and open sets.

2.7 Convexity

Every norm ball is convex by the triangle inequality.

2.8 Finite versus infinite dimensions

All norms are equivalent in finite dimensions, but this can fail in infinite-dimensional spaces.

2.9 Core structural results

Theorem 2.1 (Cauchy–Schwarz). *For $x, y \in \mathbb{R}^n$, $|\langle x, y\rangle| \le \|x\|_2\|y\|_2$.*

Proof. Apply nonnegativity to $\|x - ty\|_2^2$ and require its quadratic discriminant to be nonpositive. □

Theorem 2.2 (Euclidean triangle inequality). $\|x + y\|_2 \le \|x\|_2 + \|y\|_2$.

Proof. Expand the square and use Cauchy–Schwarz on the cross term. □

Theorem 2.3 (Norm comparison). *On $\mathbb{R}^n$, $\|x\|_\infty \le \|x\|_2 \le \sqrt{n}\|x\|_\infty$.*

Proof. The largest coordinate square is bounded by the sum of squares, and each coordinate square is bounded by the largest one. □

2.10 Worked examples

Example 2.4 (Compare 1- and 2-norms). Prove $\|x\|_2 \le \|x\|_1 \le \sqrt{n}\|x\|_2$. The first follows by expanding $(\sum |x_i|)^2$; the second is Cauchy–Schwarz against $(1, \dots, 1)$.

Example 2.5 (A seminorm, not a norm). Is $p(x, y) = |x - y|$ a norm on $\mathbb{R}^2$? No: $p(1, 1) = 0$ although $(1, 1) \ne 0$.

Example 2.6 (Weighted Euclidean norm). Show $\sqrt{4x^2 + y^2}$ is a norm. It is the Euclidean norm after the invertible linear change $(x, y) \mapsto (2x, y)$.

Example 2.7 (Distance to a coordinate axis). Find the Euclidean distance from $(1, 2)$ to the x-axis. Minimizing $(1 - t)^2 + 4$ over $(t, 0)$ gives $t = 1$ and distance 2.

2.11 Solved dossiers

Each dossier is a substantial solved problem. Corpus mappings guide topic choice where explicit retained source blocks exist; otherwise the dossier is a fresh canonical completion.

Problem 2.8 (Compare 1- and 2-norms). Prove $\|x\|_2 \le \|x\|_1 \le \sqrt{n}\|x\|_2$.

Solution. The first follows by expanding $(\sum |x_i|)^2$; the second is Cauchy–Schwarz against $(1, \dots, 1)$. □

Problem 2.9 (A seminorm, not a norm). Is $p(x, y) = |x - y|$ a norm on $\mathbb{R}^2$?

Solution. No: $p(1,1) = 0$ although $(1,1) \neq 0$. □

Problem 2.10 (Weighted Euclidean norm)**.** Show $\sqrt{4x^2 + y^2}$ is a norm.

Solution. It is the Euclidean norm after the invertible linear change $(x, y) \mapsto (2x, y)$. □

Problem 2.11 (Distance to a coordinate axis)**.** Find the Euclidean distance from $(1,2)$ to the x-axis.

Solution. Minimizing $(1-t)^2 + 4$ over $(t, 0)$ gives $t = 1$ and distance 2. □

Problem 2.12 (Projection operator norm)**.** For $P(x, y) = (x, 0)$, find $\|P\|_2$.

Solution. $\|P(x, y)\| \leq \|(x, y)\|$ and equality holds at $(1,0)$, so $\|P\| = 1$. □

Problem 2.13 (Equivalent convergence)**.** Show $\|x_n - x\|_2 \to 0$ iff $\|x_n - x\|_\infty \to 0$.

Solution. Use $\|z\|_\infty \leq \|z\|_2 \leq \sqrt{n}\|z\|_\infty$. □

Problem 2.14 (Diameter of the unit sphere)**.** If $\|x\| = \|y\| = 1$, show $\|x - y\| \leq 2$.

Solution. Use the triangle inequality on $x + (-y)$. □

Problem 2.15 (Reverse triangle inequality)**.** Prove $|\|x\| - \|y\|| \leq \|x - y\|$.

Solution. Apply the triangle inequality to $x = (x - y) + y$ and then swap x, y. □

Problem 2.16 (Norm of a Euclidean functional)**.** For $f(x) = a \cdot x$, compute $\|f\|$.

Solution. Cauchy–Schwarz gives $\|f\| \leq \|a\|_2$, with equality at $x = a/\|a\|_2$. □

Problem 2.17 (Convexity of closed balls)**.** Prove every closed norm ball is convex.

Solution. A convex combination of two radius-r displacement vectors has norm at most the same convex combination of their norms, hence at most r. □

Problem 2.18 (Infinite-dimensional warning)**.** Why is there no C with $\|f\|_\infty \leq C\|f\|_1$ for every continuous f on $[0,1]$?

Solution. Continuous spike functions can keep height 1 while their support and integral shrink to zero. □

Problem 2.19 (Equivalent-norm synthesis)**.** Why do two-sided norm bounds imply identical topology?

Solution. Each ball for one norm contains a suitably scaled ball for the other, in both directions. □

2.12 Exercises with complete solutions

The shorter exercise layer is kept separate from the dossiers.

Exercise 2.20. Compute all three standard norms of $(1, -1, 2)$.

Hint. Use definitions. □

Solution. They are $4, \sqrt{6}, 2$. □

Exercise 2.21. Show $2|x| + 3|y|$ is a norm.

Hint. Check the axioms. □

Solution. Positive weighted sums of absolute values satisfy the norm axioms. □

Exercise 2.22. Is $\max(|x|, 2|y|)$ a norm?

Hint. View it as a weighted infinity norm. □

Solution. Yes. □

Exercise 2.23. Find the Euclidean distance from $(1, 2)$ to $(-2, 6)$.

Hint. Subtract first. □

Solution. The difference has length 5. □

Exercise 2.24. Prove $\|0\| = 0$.

Hint. Use homogeneity with scalar zero. □

Solution. $\|0x\| = 0\|x\| = 0$. □

Exercise 2.25. If $x_n \to x$, show $\|x_n\| \to \|x\|$.

Hint. Use reverse triangle inequality. □

Solution. The norm difference is bounded by $\|x_n - x\|$. □

Exercise 2.26. Find the operator norm of $T(x, y) = (3x, 0)$.

Hint. Bound and attain. □

Solution. The norm is 3. □

Exercise 2.27. Show the open unit ball is convex.

Hint. Use strict norm bounds. □

Solution. A convex combination has norm below the corresponding convex combination of two numbers below 1. □

Chapter summary

The central definitions, estimates, and structural theorems of this chapter will be used repeatedly in the remaining analysis volume.

Chapter 3

Sequences and Cauchy Sequences

Sequences encode limiting behavior through quantifiers. Cauchy sequences compare late terms only with each other, and completeness is exactly the principle that such internal coherence produces a limit.

Learning goals

The reader should be able to use the chapter's definitions in proofs, recognize the decisive hypotheses, and move between concrete metric examples and invariant topological statements.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{structural theorem} \longrightarrow \text{application.}}$$

3.1 Convergence

A sequence converges when all sufficiently late terms lie in every prescribed neighborhood of a candidate limit.

3.2 Uniqueness

Metric separation makes limits unique.

3.3 Boundedness

Every convergent sequence is bounded, but boundedness alone does not force convergence.

3.4 Subsequences

Subsequences preserve order and are the basic tool for detecting hidden limiting behavior.

3.5 Cauchy sequences

A sequence is Cauchy when sufficiently late terms are mutually close.

3.6 Completeness

A metric space is complete if every Cauchy sequence converges in the space.

3.7 Monotone real sequences

Bounded monotone real sequences converge by the supremum property.

3.8 Limsup and liminf

Tail suprema and infima capture asymptotic oscillation.

3.9 Core structural results

Theorem 3.1 (Convergent implies Cauchy)**.** *Every convergent sequence in a metric space is Cauchy.*

Proof. Choose a tail within $\varepsilon/2$ of the limit and apply the triangle inequality. □

Theorem 3.2 (Monotone convergence)**.** *Every increasing real sequence bounded above converges to the supremum of its range.*

Proof. Supremum approximation supplies one late term within ε below the supremum; monotonicity traps every later term. □

Theorem 3.3 (Cauchy boundedness)**.** *Every Cauchy sequence is bounded.*

Proof. A tail lies in a unit ball about one tail term; combine this with the finite initial segment. □

3.10 Worked examples

Example 3.4 (Direct proof for $1/n$)**.** Prove $1/n \to 0$. Given $\varepsilon > 0$, choose $N > 1/\varepsilon$.

Example 3.5 (Uniqueness of limits)**.** Prove a metric sequence cannot converge to two distinct points. The triangle inequality would make the distance between the two proposed limits smaller than itself.

Example 3.6 (Convergent implies bounded)**.** Prove it without invoking a theorem name. Take a tail in a unit ball around the limit and enlarge the ball to contain the finite initial segment.

Example 3.7 (Subsequence inheritance)**.** Show every subsequence of a convergent sequence has the same limit. Subsequence indices eventually exceed any original convergence index.

3.11 Solved dossiers

Each dossier is a substantial solved problem. Corpus mappings guide topic choice where explicit retained source blocks exist; otherwise the dossier is a fresh canonical completion.

Problem 3.8 (Direct proof for $1/n$)**.** Prove $1/n \to 0$.

Solution. Given $\varepsilon > 0$, choose $N > 1/\varepsilon$. □

Problem 3.9 (Uniqueness of limits)**.** Prove a metric sequence cannot converge to two distinct points.

Solution. The triangle inequality would make the distance between the two proposed limits smaller than itself. □

Problem 3.10 (Convergent implies bounded)**.** Prove it without invoking a theorem name.

Solution. Take a tail in a unit ball around the limit and enlarge the ball to contain the finite initial segment. □

Problem 3.11 (Subsequence inheritance)**.** Show every subsequence of a convergent sequence has the same limit.

Solution. Subsequence indices eventually exceed any original convergence index. □

Problem 3.12 (A Cauchy sequence in $\mathbb{Q}$ without rational limit)**.** Use rational truncations of $\sqrt{2}$.

Solution. They converge in $\mathbb{R}$ and are therefore Cauchy, but a rational limit would equal $\sqrt{2}$ by uniqueness. □

Problem 3.13 (Geometric-series Cauchy criterion)**.** Show partial sums of $\sum r^k$ are Cauchy for $|r| < 1$.

Solution. Bound every tail by $|r|^{n+1}/(1 - |r|)$. □

Problem 3.14 (Recursive monotone convergence)**.** For $x_{n+1} = \sqrt{2 + x_n}$, $x_1 = 1$, prove $x_n \to 2$.

Solution. Inductively the sequence is increasing and bounded by 2; its limit satisfies $L^2 = L+2$ and positivity forces $L = 2$. □

Problem 3.15 (Bounded but divergent)**.** Analyze $(-1)^n$.

Solution. Even and odd subsequences converge to different values, so the full sequence diverges. □

Problem 3.16 (Limsup and liminf)**.** Find them for $(-1)^n + 1/n$.

Solution. Even and odd subsequences approach 1 and -1, giving limsup 1 and liminf -1. □

Problem 3.17 (Cauchy plus a convergent subsequence)**.** Prove the whole sequence converges to the subsequential limit.

Solution. Use one late subsequence term as a bridge between the Cauchy tail and the limit. □

Problem 3.18 (Completeness of reals via tail bounds)**.** Explain how a Cauchy real sequence can be trapped by tail infima and suprema.

Solution. Tail infima increase, tail suprema decrease, and their gap tends to zero; completeness gives a common limiting value. □

Problem 3.19 (Sequence synthesis)**.** Distinguish convergence from the Cauchy property conceptually.

Solution. Convergence references an external limit; Cauchy behavior is internal. Completeness connects the two. □

3.12 Exercises with complete solutions

The shorter exercise layer is kept separate from the dossiers.

Exercise 3.20. Prove $3/n \to 0$.

Hint. Choose $N > 3/\varepsilon$. □

Solution. Then $n \geq N$ implies $3/n < \varepsilon$. □

Exercise 3.21. Does $(-1)^n/n$ converge?

Hint. Bound the absolute value. □

Solution. Yes, to 0. □

Exercise 3.22. Show every constant sequence is Cauchy.

Hint. Distances vanish. □

Solution. Every pairwise distance is zero. □

Exercise 3.23. If $x_n \to x$, prove $x_{n+1} \to x$.

Hint. It is a tail subsequence. □

Solution. The same epsilon estimate applies. □

Exercise 3.24. Find $\lim n/(n+1)$.

Hint. Rewrite the expression. □

Solution. It is 1. □

Exercise 3.25. Give a bounded divergent sequence.

Hint. Use alternating signs. □

Solution. $(-1)^n$ works. □

Exercise 3.26. What happens to an increasing bounded real sequence?

Hint. Use monotone convergence. □

Solution. It converges to the supremum of its range. □

Exercise 3.27. Show a Cauchy tail is bounded.

Hint. Use epsilon 1. □

Solution. Eventually all terms lie in a unit ball around one fixed tail term. □

Chapter summary

The central definitions, estimates, and structural theorems of this chapter will be used repeatedly in the remaining analysis volume.

Chapter 4

Open and Closed Sets

Open and closed sets encode local and limiting structure without coordinates. Interiors, closures, boundaries, and limit points become the language in which continuity and topology are most naturally expressed.

Learning goals

The reader should be able to use the chapter's definitions in proofs, recognize the decisive hypotheses, and move between concrete metric examples and invariant topological statements.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{structural theorem} \longrightarrow \text{application.}}$$

4.1 Open sets

A set is open if every one of its points contains a ball lying entirely in the set.

4.2 Closed sets

A set is closed if its complement is open; in metric spaces this is equivalent to containing limits of convergent sequences from the set.

4.3 Interior

The interior is the largest open subset contained in a set.

4.4 Closure

The closure is the smallest closed set containing the set.

4.5 Limit points

A point is a limit point if every punctured neighborhood meets the set.

4.6 Boundary

Boundary points have every neighborhood meeting both the set and its complement.

4.7 Set operations

Arbitrary unions and finite intersections preserve openness; dual statements hold for closed sets.

4.8 Relative topology and density

Relative openness is obtained by intersecting ambient open sets with a subspace, while density means the closure fills the ambient space.

4.9 Core structural results

Theorem 4.1 (Open-set operations)**.** *Arbitrary unions and finite intersections of open sets are open.*

Proof. For unions inherit a ball from one containing member; for finite intersections take the minimum of finitely many available radii. □

Theorem 4.2 (Sequential closedness)**.** *A metric subset is closed iff it contains every limit of every convergent sequence drawn from it.*

Proof. Closedness prevents a convergent sequence from entering an open complement; conversely a point in the closure but outside the set yields points within $1/n$ converging to it. □

Theorem 4.3 (Closure by balls)**.** $x \in \overline{A}$ *iff every ball centered at* x *meets* A*.*

Proof. A ball missing A places x in the open complement of the closure; the converse is immediate from the same characterization. □

4.10 Worked examples

Example 4.4 (Interval anatomy)**.** Find the interior, closure, and boundary of $(0, 1]$. They are $(0, 1)$, $[0, 1]$, and $\{0, 1\}$.

Example 4.5 (Open balls are open)**.** Prove every open metric ball is open. For $x \in B(a, r)$ choose radius $r - d(a, x) > 0$ and use the triangle inequality.

Example 4.6 (Closed balls are closed)**.** Prove every closed metric ball is closed. If x_n lies in the ball and converges to x, continuity of $y \mapsto d(a, y)$ gives $d(a, x) \leq r$.

Example 4.7 (Sphere is closed)**.** Show $\{x : \|x\| = 1\}$ is closed in a normed space. The norm is continuous by the reverse triangle inequality, so the sphere is the inverse image of the closed singleton $\{1\}$.

4.11 Solved dossiers

Each dossier is a substantial solved problem. Corpus mappings guide topic choice where explicit retained source blocks exist; otherwise the dossier is a fresh canonical completion.

Problem 4.8 (Interval anatomy)**.** Find the interior, closure, and boundary of $(0, 1]$.

Solution. They are $(0, 1)$, $[0, 1]$, and $\{0, 1\}$. □

Problem 4.9 (Open balls are open)**.** Prove every open metric ball is open.

Solution. For $x \in B(a, r)$ choose radius $r - d(a, x) > 0$ and use the triangle inequality. □

Problem 4.10 (Closed balls are closed)**.** Prove every closed metric ball is closed.

Solution. If x_n lies in the ball and converges to x, continuity of $y \mapsto d(a, y)$ gives $d(a, x) \leq r$. □

Problem 4.11 (Sphere is closed)**.** Show $\{x : \|x\| = 1\}$ is closed in a normed space.

Solution. The norm is continuous by the reverse triangle inequality, so the sphere is the inverse image of the closed singleton $\{1\}$. □

Problem 4.12 (Boundary criterion)**.** Show $x \in \partial A$ iff every ball about x meets both A and its complement.

Solution. Membership in the closure gives intersection with A; failure to be interior gives intersection with the complement. □

Problem 4.13 (Closure of rationals)**.** Prove $\overline{\mathbb{Q}} = \mathbb{R}$.

Solution. Rational density says every real ball contains a rational point. □

Problem 4.14 (Interior of rationals)**.** Show $\mathbb{Q}^\circ = \varnothing$.

Solution. Every interval around a rational contains irrational points. □

Problem 4.15 (A clopen subspace piece)**.** In $Y = (-\infty, 0) \cup (0, \infty)$, show $(0, \infty)$ is clopen relative to Y.

Solution. It and its complement are intersections of Y with ambient open half-lines. □

Problem 4.16 (Closure of a finite union)**.** Prove $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

Solution. The right side is closed and contains the union; monotonicity of closure gives the reverse inclusion. □

Problem 4.17 (Interior of an intersection)**.** Prove $(A \cap B)^\circ = A^\circ \cap B^\circ$.

Solution. The right side is open and contained in the intersection; every open subset of the intersection lies in both interiors. □

Problem 4.18 (Limit-point sequence)**.** If x is a limit point of A in a metric space, construct a sequence of points of $A \setminus \{x\}$ converging to x.

Solution. Choose $x_n \in A \cap (B(x, 1/n) \setminus \{x\})$; then $d(x_n, x) < 1/n$. □

Problem 4.19 (Topology synthesis)**.** Why can topology be specified by open sets without mentioning a metric?

Solution. Convergence, closedness, continuity, compactness, connectedness, interior, closure, and boundary can all be defined from the family of open sets. □

4.12 Exercises with complete solutions

The shorter exercise layer is kept separate from the dossiers.

Exercise 4.20. Is $(0,1)$ open in $\mathbb{R}$?

Hint. Choose a point-dependent radius. □

Solution. Yes. □

Exercise 4.21. Is $[0,1]$ closed?

Hint. Inspect the complement. □

Solution. Yes. □

Exercise 4.22. Find $\partial(0,1)$.

Hint. Use closure minus interior. □

Solution. It is $\{0,1\}$. □

Exercise 4.23. Find $\overline{\mathbb{Z}}$ in $\mathbb{R}$.

Hint. Is $\mathbb{Z}$ closed? □

Solution. It equals $\mathbb{Z}$. □

Exercise 4.24. Show $\varnothing$ and X are clopen.

Hint. Use complements. □

Solution. Each is open and the complement of the other. □

Exercise 4.25. Are arbitrary intersections of open sets open?

Hint. Use shrinking intervals. □

Solution. No; $\cap_n(-1/n, 1/n) = \{0\}$. □

Exercise 4.26. Find $[0,1]^\circ$.

Hint. Endpoints fail the ball condition. □

Solution. It is $(0,1)$. □

Exercise 4.27. Show $A^\circ \subset A \subset \overline{A}$.

Hint. Use the defining extremal properties. □

Solution. Both inclusions are immediate from the definitions. □

Chapter summary

The central definitions, estimates, and structural theorems of this chapter will be used repeatedly in the remaining analysis volume.

Chapter 5

Metric Spaces and Continuity

Metric spaces isolate the role of distance in analysis. Continuity becomes preservation of local closeness, with equivalent epsilon-delta, sequential, and open-set formulations.

Learning goals

The reader should be able to use the chapter's definitions in proofs, recognize the decisive hypotheses, and move between concrete metric examples and invariant topological statements.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{structural theorem} \longrightarrow \text{application.}}$$

5.1 Metrics

A metric is nonnegative, definite, symmetric, and satisfies the triangle inequality.

5.2 Metric examples

Norm metrics, discrete metrics, and bounded transforms of metrics show how one set can carry different geometries.

5.3 Metric convergence

A sequence converges exactly when its distance to the limit tends to zero.

5.4 Continuity at a point

Output tolerances are controlled by sufficiently small input tolerances.

5.5 Sequential continuity

In metric spaces continuity is equivalent to preserving sequential limits.

5.6 Open-set continuity

Continuity is equivalent to inverse images of open sets being open.

5.7 Uniform continuity

One input scale must work simultaneously at every point.

5.8 Lipschitz maps

A Lipschitz estimate gives quantitative control and automatically implies uniform continuity.

5.9 Core structural results

Theorem 5.1 (Sequential criterion)**.** *A map between metric spaces is continuous iff it preserves limits of sequences.*

Proof. Continuity transfers epsilon estimates to a convergent sequence. Failure of continuity creates points within $1/n$ whose images stay a fixed distance away. ☐

Theorem 5.2 (Open-set criterion)**.** *A map is continuous iff the inverse image of every open set is open.*

Proof. At a point in a preimage, continuity pulls back a small target ball; conversely apply openness to balls around the image point. ☐

Theorem 5.3 (Lipschitz implies uniform continuity)**.** *If $d(f(x), f(y)) \leq Ld(x, y)$, then f is uniformly continuous.*

Proof. For $L > 0$ choose $\delta = \varepsilon/L$; for $L = 0$ the map is constant. ☐

5.10 Worked examples

Example 5.4 (Discrete metric topology)**.** Show every subset is open in the discrete metric. For any point x, the ball $B(x, 1/2)$ is the singleton $\{x\}$.

Example 5.5 (Norm gives a metric)**.** Prove $d(x, y) = \|x - y\|$ is a metric. Definiteness and symmetry come from norm axioms, and the metric triangle inequality is the norm triangle inequality.

Example 5.6 (Distance to a fixed point is Lipschitz)**.** Show $x \mapsto d(x, a)$ is 1-Lipschitz. The triangle inequality gives $|d(x, a) - d(y, a)| \leq d(x, y)$.

Example 5.7 (A sequential discontinuity witness)**.** Define $f(0) = 0$ and $f(x) = 1/x$ otherwise. Show discontinuity at zero. Take $x_n = 1/n \to 0$; then $f(x_n) = n$ does not converge to 0.

5.11 Solved dossiers

Each dossier is a substantial solved problem. Corpus mappings guide topic choice where explicit retained source blocks exist; otherwise the dossier is a fresh canonical completion.

Problem 5.8 (Discrete metric topology)**.** Show every subset is open in the discrete metric.

Solution. For any point x, the ball $B(x, 1/2)$ is the singleton $\{x\}$. □

Problem 5.9 (Norm gives a metric)**.** Prove $d(x, y) = \|x - y\|$ is a metric.

Solution. Definiteness and symmetry come from norm axioms, and the metric triangle inequality is the norm triangle inequality. □

Problem 5.10 (Distance to a fixed point is Lipschitz)**.** Show $x \mapsto d(x, a)$ is 1-Lipschitz.

Solution. The triangle inequality gives $|d(x, a) - d(y, a)| \leq d(x, y)$. □

Problem 5.11 (A sequential discontinuity witness)**.** Define $f(0) = 0$ and $f(x) = 1/x$ otherwise. Show discontinuity at zero.

Solution. Take $x_n = 1/n \to 0$; then $f(x_n) = n$ does not converge to 0. □

Problem 5.12 (Square is not uniformly continuous on $\mathbb{R}$)**.** Produce nearby points with separated squares.

Solution. Take $x_n = n$, $y_n = n + 1/n$; input distance tends to zero while square difference tends to 2. □

Problem 5.13 (Square on a bounded interval)**.** Show x^2 is uniformly continuous on $[-M, M]$.

Solution. $|x^2 - y^2| \leq 2M|x - y|$, so it is Lipschitz there. □

Problem 5.14 (Composition)**.** Prove the composition of continuous maps is continuous.

Solution. Inverse images compose: $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$. □

Problem 5.15 (A bounded equivalent metric)**.** Show $\rho = d/(1 + d)$ has the same convergent sequences as d.

Solution. $\rho \leq d$, while for small ρ, $d = \rho/(1 - \rho) \leq 2\rho$. □

Problem 5.16 (Closed-set criterion)**.** Show continuous maps pull closed sets back to closed sets.

Solution. Take complements and use the open-set criterion. □

Problem 5.17 (Uniform continuity preserves Cauchy sequences)**.** Prove the claim.

Solution. A uniform delta converts a Cauchy input tail into an epsilon-close image tail. □

Problem 5.18 (Lipschitz diameter control)**.** Show $\operatorname{diam} f(A) \leq L \operatorname{diam} A$.

Solution. Apply the Lipschitz inequality to every pair and take the supremum. □

Problem 5.19 (Continuity synthesis)**.** Why do inverse images appear in the topological definition?

Solution. Inverse images preserve arbitrary unions and finite intersections exactly, matching the axioms for open sets. □

5.12 Exercises with complete solutions

The shorter exercise layer is kept separate from the dossiers.

Exercise 5.20. Verify the triangle inequality for the discrete metric.

Hint. Consider whether endpoints agree. □

Solution. If endpoints differ, at least one leg has length 1. □

Exercise 5.21. Is $f(x) = 3x$ Lipschitz?

Hint. Compute differences. □

Solution. Yes, with constant 3. □

Exercise 5.22. Show a constant map is uniformly continuous.

Hint. Output distance is zero. □

Solution. Any positive delta works. □

Exercise 5.23. Are sums of continuous real functions continuous?

Hint. Use limits or epsilon bounds. □

Solution. Yes. □

Exercise 5.24. Show $|x|$ is continuous.

Hint. Use reverse triangle inequality. □

Solution. It is 1-Lipschitz. □

Exercise 5.25. Characterize convergence in the discrete metric.

Hint. Use epsilon 1/2. □

Solution. A convergent sequence is eventually constant. □

Exercise 5.26. Does uniform continuity imply continuity?

Hint. Fix a point. □

Solution. Yes. □

Exercise 5.27. Give a continuous but not uniformly continuous map on $\mathbb{R}$.

Hint. Use quadratic growth. □

Solution. x^2 is a standard example. □

Chapter summary

The central definitions, estimates, and structural theorems of this chapter will be used repeatedly in the remaining analysis volume.

Chapter 6

Compactness

Compactness is the finiteness principle of analysis. It converts infinite covers into finite subcovers, arbitrary sequences into convergent subsequences, and local continuity into global boundedness and uniform control.

Learning goals

The reader should be able to use the chapter's definitions in proofs, recognize the decisive hypotheses, and move between concrete metric examples and invariant topological statements.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{structural theorem} \longrightarrow \text{application.}}$$

6.1 Open-cover compactness

Every open cover of a compact space has a finite subcover.

6.2 Sequential compactness

In metric spaces compactness is equivalent to every sequence having a convergent subsequence with limit in the space.

6.3 Heine–Borel

In Euclidean space compact sets are exactly the closed and bounded sets.

6.4 Bolzano–Weierstrass

Every bounded sequence in Euclidean space has a convergent subsequence.

6.5 Continuous images

Continuous images of compact spaces are compact.

6.6 Extreme values

Continuous real-valued functions on compact sets attain maxima and minima.

6.7 Uniform continuity

Continuous maps on compact metric domains are uniformly continuous.

6.8 Completeness and total boundedness

A metric space is compact exactly when it is complete and totally bounded.

6.9 Core structural results

Theorem 6.1 (Continuous image theorem)**.** *If K is compact and f is continuous, then $f(K)$ is compact.*

Proof. Pull an open cover of the image back to K, extract a finite subcover, and push the corresponding finite family forward. ☐

Theorem 6.2 (Extreme-value theorem)**.** *A continuous real function on nonempty compact K attains its maximum and minimum.*

Proof. Its image is compact in $\mathbb{R}$, hence closed and bounded; its supremum and infimum belong to the image. ☐

Theorem 6.3 (Heine–Cantor)**.** *A continuous map on a compact metric domain is uniformly continuous.*

Proof. Otherwise choose pairs arbitrarily close with images separated by a fixed epsilon; a convergent subsequence of one side forces both sides to the same limit, contradicting continuity. ☐

6.10 Worked examples

Example 6.4 (Open interval is not compact)**.** Use an open cover to show $(0,1)$ is not compact. The sets $(1/n,1)$ for $n \geq 2$ cover $(0,1)$ but no finite subfamily covers points near zero.

Example 6.5 (Sequential noncompactness)**.** Show $(0,1]$ is not compact using a sequence. The sequence $1/n$ has no subsequence converging to a point of $(0,1]$.

Example 6.6 (Compact sets are closed)**.** Prove this in metric spaces. For $x \notin K$, the continuous distance to x attains a positive minimum on compact K, leaving a ball around x disjoint from K.

Example 6.7 (Compact sets are bounded)**.** Prove it directly from an open cover. The increasing balls $B(x_0,n)$ cover the space; finitely many reduce to one largest ball.

6.11 Solved dossiers

Each dossier is a substantial solved problem. Corpus mappings guide topic choice where explicit retained source blocks exist; otherwise the dossier is a fresh canonical completion.

Problem 6.8 (Open interval is not compact)**.** Use an open cover to show $(0, 1)$ is not compact.

Solution. The sets $(1/n, 1)$ for $n \geq 2$ cover $(0, 1)$ but no finite subfamily covers points near zero. □

Problem 6.9 (Sequential noncompactness)**.** Show $(0, 1]$ is not compact using a sequence.

Solution. The sequence $1/n$ has no subsequence converging to a point of $(0, 1]$. □

Problem 6.10 (Compact sets are closed)**.** Prove this in metric spaces.

Solution. For $x \notin K$, the continuous distance to x attains a positive minimum on compact K, leaving a ball around x disjoint from K. □

Problem 6.11 (Compact sets are bounded)**.** Prove it directly from an open cover.

Solution. The increasing balls $B(x_0, n)$ cover the space; finitely many reduce to one largest ball. □

Problem 6.12 (Continuous real functions are bounded)**.** Prove boundedness on compact domains.

Solution. The compact image in $\mathbb{R}$ is bounded. □

Problem 6.13 (Positive distance between disjoint compact sets)**.** Prove disjoint compact K, L have positive distance.

Solution. The continuous function $d(x, y)$ on compact $K \times L$ attains a minimum; zero would force a common point. □

Problem 6.14 (Finite intersection property)**.** Prove compact spaces have it for closed sets.

Solution. If the total closed intersection were empty, their complements would give an open cover with a finite subcover, contradicting the finite intersection hypothesis. □

Problem 6.15 (Nested compact sets)**.** Show decreasing nonempty compact sets have nonempty intersection.

Solution. They satisfy the finite intersection property inside the first compact set. □

Problem 6.16 (Heine–Cantor contradiction sequence)**.** Describe the standard sequence proof.

Solution. Failure gives $d(x_n, y_n) < 1/n$ but image distance at least ε_0; compactness and continuity contradict this. □

Problem 6.17 (Bolzano–Weierstrass by nested boxes)**.** Sketch the construction.

Solution. Repeatedly choose a smaller closed box containing infinitely many sequence terms and diameters tending to zero; choose one term from each stage. □

Problem 6.18 (Closed subset of compact is compact)**.** Prove it by covers.

Solution. Add the open complement of the closed subset to any cover, extract a finite cover of the ambient compact set, then discard the complement. ☐

Problem 6.19 (Compactness synthesis)**.** Why is compactness a finite-extraction principle?

Solution. Its cover definition extracts finitely many neighborhoods, and its sequential form extracts one convergent subsequence from infinitely many terms. ☐

6.12 Exercises with complete solutions

The shorter exercise layer is kept separate from the dossiers.

Exercise 6.20. Is $\{1/n : n \geq 1\} \cup \{0\}$ compact?

Hint. Check closed and bounded. ☐

Solution. Yes. ☐

Exercise 6.21. Is $\mathbb{Z}$ compact in $\mathbb{R}$?

Hint. Check boundedness. ☐

Solution. No. ☐

Exercise 6.22. Show every finite metric space is compact.

Hint. Choose one cover member per point. ☐

Solution. Finitely many choices suffice. ☐

Exercise 6.23. Does compactness imply completeness in metric spaces?

Hint. Use a convergent subsequence of a Cauchy sequence. ☐

Solution. Yes. ☐

Exercise 6.24. Is $(0, 1)$ compact?

Hint. Use Heine–Borel. ☐

Solution. No. ☐

Exercise 6.25. Is a continuous image of compact closed in a metric codomain?

Hint. Compact subsets of metric spaces are closed. ☐

Solution. Yes. ☐

Exercise 6.26. Why is $[a, b]$ compact?

Hint. Use Heine–Borel. ☐

Solution. It is closed and bounded. ☐

Exercise 6.27. Must every sequence in a compact set converge?

Hint. Only a subsequence is guaranteed. ☐

Solution. No; $(-1)^n$ in $\{-1, 1\}$ is a counterexample. ☐

Chapter summary

The central definitions, estimates, and structural theorems of this chapter will be used repeatedly in the remaining analysis volume.

Chapter 7

Connectedness and Path Connectedness

Connectedness excludes decompositions into separated open pieces, while path connectedness asks for explicit continuous curves between points. These notions capture global cohesion that compactness and completeness do not.

Learning goals

The reader should be able to use the chapter's definitions in proofs, recognize the decisive hypotheses, and move between concrete metric examples and invariant topological statements.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{structural theorem} \longrightarrow \text{application.}}$$

7.1 Connected spaces

A space is connected if it admits no separation into disjoint nonempty open pieces.

7.2 Clopen criterion

Connected spaces have no nontrivial subset that is both open and closed.

7.3 Connected subsets of the line

A subset of $\mathbb{R}$ is connected exactly when it is an interval in the order-convex sense.

7.4 Continuous images

Continuous maps preserve connectedness.

7.5 Intermediate values

The intermediate-value theorem follows because continuous images of intervals are intervals.

7.6 Path connectedness

A path is a continuous map from $[0, 1]$ joining two prescribed endpoints.

7.7 Path connected implies connected

A path cannot cross a separation without separating the interval parameter.

7.8 Components

Connected and path components are maximal subsets with the corresponding cohesion property.

7.9 Core structural results

Theorem 7.1 (Continuous image theorem). *The continuous image of a connected space is connected.*

Proof. A separation of the image pulls back to a separation of the domain. □

Theorem 7.2 (Intervals are connected). *Every real interval is connected.*

Proof. A hypothetical separation and a supremum argument at the interface produce a point that cannot consistently belong to either relatively open side. □

Theorem 7.3 (Path connected implies connected). *Every path-connected space is connected.*

Proof. A separation of the space would pull back along a path joining opposite sides to a separation of $[0, 1]$. □

7.10 Worked examples

Example 7.4 (Clopen criterion). Prove connectedness is equivalent to having only trivial clopen subsets. A nontrivial clopen set and its complement form a separation, and every separation makes each side clopen.

Example 7.5 (Punctured line). Show $\mathbb{R} \setminus \{0\}$ is disconnected. The negative and positive half-lines are disjoint nonempty relatively open sets covering it.

Example 7.6 (Connected subsets of the line). If $A \subset \mathbb{R}$ is connected and $x < z < y$ with $x, y \in A$, show $z \in A$. If $z \notin A$, split A at z into points below and above, creating a separation.

Example 7.7 (Intermediate-value theorem). Derive it from connectedness. The continuous image of $[a, b]$ is connected in $\mathbb{R}$, hence an interval containing all values between endpoint values.

7.11 Solved dossiers

Each dossier is a substantial solved problem. Corpus mappings guide topic choice where explicit retained source blocks exist; otherwise the dossier is a fresh canonical completion.

Problem 7.8 (Clopen criterion)**.** Prove connectedness is equivalent to having only trivial clopen subsets.

Solution. A nontrivial clopen set and its complement form a separation, and every separation makes each side clopen. ☐

Problem 7.9 (Punctured line)**.** Show $\mathbb{R} \setminus \{0\}$ is disconnected.

Solution. The negative and positive half-lines are disjoint nonempty relatively open sets covering it. ☐

Problem 7.10 (Connected subsets of the line)**.** If $A \subset \mathbb{R}$ is connected and $x < z < y$ with $x, y \in A$, show $z \in A$.

Solution. If $z \notin A$, split A at z into points below and above, creating a separation. ☐

Problem 7.11 (Intermediate-value theorem)**.** Derive it from connectedness.

Solution. The continuous image of $[a, b]$ is connected in $\mathbb{R}$, hence an interval containing all values between endpoint values. ☐

Problem 7.12 (Convex implies path connected)**.** Prove every convex subset of a normed vector space is path connected.

Solution. The straight-line path $(1 - t)x + ty$ remains in the convex set. ☐

Problem 7.13 (Path components are disjoint)**.** Show two path components are equal or disjoint.

Solution. If they meet, concatenate paths through a common point to show their union is path connected, contradicting maximality unless they coincide. ☐

Problem 7.14 (Union through a common point)**.** Show a union of connected sets with a common point is connected.

Solution. Any separation would place the common point on one side; each connected member meeting that side must lie entirely there. ☐

Problem 7.15 (Closure preserves connectedness)**.** Prove the closure of a connected set is connected.

Solution. A separation of the closure would meet the original dense connected set on both sides and induce a separation of it. ☐

Problem 7.16 (Continuous image of a path)**.** Show continuous maps preserve path connectedness.

Solution. Compose the original path with the continuous map. ☐

Problem 7.17 (Components are closed)**.** Prove every connected component is closed.

Solution. The closure of a connected set is connected and contains the component; maximality forces equality. ☐

Problem 7.18 (Products of path-connected spaces)**.** Show $X \times Y$ is path connected when both factors are.

Solution. Pair paths coordinatewise: $t \mapsto (\alpha(t), \beta(t))$. ☐

Problem 7.19 (Connectedness synthesis)**.** Contrast connectedness with compactness.

Solution. Connectedness excludes a global split into separated regions; compactness excludes infinite escape by finite extraction. Neither implies the other in general. ☐

7.12 Exercises with complete solutions

The shorter exercise layer is kept separate from the dossiers.

Exercise 7.20. Is $[0, 1] \cup [2, 3]$ connected?

Hint. Find a separation. ☐

Solution. No. ☐

Exercise 7.21. Is $\mathbb{R}$ path connected?

Hint. Use straight lines. ☐

Solution. Yes. ☐

Exercise 7.22. Are continuous images of connected sets connected?

Hint. Use the theorem. ☐

Solution. Yes. ☐

Exercise 7.23. Are connected subsets of $\mathbb{R}$ intervals?

Hint. Use an intermediate-point argument. ☐

Solution. Yes. ☐

Exercise 7.24. Show a singleton is connected.

Hint. There is no nontrivial split. ☐

Solution. Its only clopen subsets are trivial. ☐

Exercise 7.25. Is $\mathbb{Q}$ connected?

Hint. Cut at an irrational number. ☐

Solution. No. ☐

Exercise 7.26. Does path connected imply connected?

Hint. Use interval connectedness. ☐

Solution. Yes. ☐

Exercise 7.27. Does connected imply path connected?

Hint. Recall a classical plane counterexample. ☐

Solution. No; the topologist's sine curve is a standard counterexample. ☐

Chapter summary

The central definitions, estimates, and structural theorems of this chapter will be used repeatedly in the remaining analysis volume.

Part II

Calculus

Chapter 8

Differentiability in Several Variables

Multivariable differentiability replaces a one-dimensional slope by the best linear approximation. The derivative is therefore an operator, not merely a list of partial derivatives, and the remainder estimate is the central invariant statement.

Learning goals

The reader should be able to use the central definitions, prove the structural results, diagnose the role of each hypothesis, and solve representative analytic problems.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

8.1 Directional and partial derivatives

Partial derivatives probe coordinate directions, while directional derivatives probe arbitrary directions; neither collection alone automatically guarantees differentiability.

8.2 Fréchet differentiability

A map $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at a when $f(a + h) = f(a) + Df(a)h + o(\|h\|)$.

8.3 Jacobian matrices

After choosing standard bases, the derivative is represented by the Jacobian matrix.

8.4 Chain rule

Derivatives compose as linear maps: $D(g \circ f)(a) = Dg(f(a))Df(a)$.

8.5 Gradient and level sets

For scalar-valued functions, $Df(a)h = \nabla f(a) \cdot h$; gradients are normal to regular level sets.

8.6 Higher derivatives

Second derivatives assemble into the Hessian, a bilinear form governing second-order behavior.

8.7 Taylor approximation

Taylor's formula organizes linear and quadratic approximations plus a controlled remainder.

8.8 Critical points

At interior extrema differentiable scalar functions have zero gradient; Hessian definiteness gives second-order classification.

8.9 Core structural results

Theorem 8.1 (Differentiability implies continuity)**.** *If f is differentiable at a, then f is continuous at a.*

Proof. Write $f(a+h) - f(a) = Df(a)h + r(h)$ with $\|r(h)\|/\|h\| \to 0$. The bounded linear term and the remainder both tend to zero. ☐

Theorem 8.2 (Chain rule)**.** *If f is differentiable at a and g at $f(a)$, then $g \circ f$ is differentiable at a with derivative $Dg(f(a))Df(a)$.*

Proof. Insert the linear-plus-remainder expansions for f and g; boundedness of Dg and the smallness of both remainders produce an $o(\|h\|)$ total error. ☐

Theorem 8.3 (Symmetry of mixed partials)**.** *If second partial derivatives are continuous near a point, then mixed partials agree there.*

Proof. Apply the one-variable mean-value theorem twice to a rectangular difference quotient and pass to the limit using continuity. ☐

8.10 Worked examples

Example 8.4 (A Jacobian by linearization)**.** For $f(x,y) = (x^2y, e^{x+y})$, compute $Df(1,0)$. The Jacobian is $\begin{pmatrix} 2xy & x^2 \\ e^{x+y} & e^{x+y} \end{pmatrix}$, hence at $(1,0)$ it is $\begin{pmatrix} 0 & 1 \\ e & e \end{pmatrix}$.

Example 8.5 (Directional derivatives without differentiability)**.** Show $f(x,y) = |xy|^{1/2}$ has directional derivatives at the origin but is not differentiable there. Along (tu, tv), $f = |t|\sqrt{|uv|}$, so one-sided behavior depends linearly only in magnitude and fails a genuine linear approximation; along $y = x$, $f(t,t) = |t|$, which is not $o(\|(t,t)\|)$.

Example 8.6 (Gradient normal to a level set)**.** For $F(x,y) = x^2 + 2y^2$, find the normal direction to $F = 3$ at $(1,1)$. $\nabla F = (2x, 4y)$, so at $(1,1)$ a normal vector is $(2,4)$.

Example 8.7 (Chain rule in coordinates)**.** Let $f(t) = (t, t^2)$ and $g(x,y) = xe^y$. Compute $(g \circ f)'(1)$ using the chain rule. $Df(1) = (1,2)^T$ and $Dg(1,1) = (e,e)$, so the derivative is $e + 2e = 3e$.

8.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 8.8 (A Jacobian by linearization)**.** For $f(x,y) = (x^2y, e^{x+y})$, compute $Df(1,0)$.

Solution. The Jacobian is $\begin{pmatrix} 2xy & x^2 \\ e^{x+y} & e^{x+y} \end{pmatrix}$, hence at $(1,0)$ it is $\begin{pmatrix} 0 & 1 \\ e & e \end{pmatrix}$. ☐

Problem 8.9 (Directional derivatives without differentiability)**.** Show $f(x,y) = |xy|^{1/2}$ has directional derivatives at the origin but is not differentiable there.

Solution. Along (tu, tv), $f = |t|\sqrt{|uv|}$, so one-sided behavior depends linearly only in magnitude and fails a genuine linear approximation; along $y = x$, $f(t,t) = |t|$, which is not $o(\|(t,t)\|)$. ☐

Problem 8.10 (Gradient normal to a level set)**.** For $F(x,y) = x^2 + 2y^2$, find the normal direction to $F = 3$ at $(1,1)$.

Solution. $\nabla F = (2x, 4y)$, so at $(1,1)$ a normal vector is $(2,4)$. ☐

Problem 8.11 (Chain rule in coordinates)**.** Let $f(t) = (t, t^2)$ and $g(x,y) = xe^y$. Compute $(g \circ f)'(1)$ using the chain rule.

Solution. $Df(1) = (1,2)^T$ and $Dg(1,1) = (e,e)$, so the derivative is $e + 2e = 3e$. ☐

Problem 8.12 (Quadratic approximation)**.** Find the Hessian of $f(x,y) = x^2 + xy + 3y^2$.

Solution. $H = \begin{pmatrix} 2 & 1 \\ 1 & 6 \end{pmatrix}$, constant everywhere. ☐

Problem 8.13 (Critical-point classification)**.** Classify the origin for $f(x,y) = x^2 + 4y^2$.

Solution. The gradient vanishes and the Hessian is positive definite, so the origin is a strict local minimum. ☐

Problem 8.14 (A saddle)**.** Classify the origin for $f(x,y) = x^2 - y^2$.

Solution. The Hessian has eigenvalues 2 and -2, hence is indefinite; the origin is a saddle. ☐

Problem 8.15 (Differentiability from continuous partials)**.** Explain why continuous first partial derivatives near a point imply differentiability there.

Solution. Move from the point to a nearby point along coordinate segments and apply the one-variable mean-value theorem; continuity of the partials controls the remainder uniformly by $o(\|h\|)$. ☐

Problem 8.16 (Derivative of a norm square)**.** For $f(x) = \|x\|_2^2$ on $\mathbb{R}^n$, compute $Df(x)$.

Solution. $Df(x)h = 2x \cdot h$, so the gradient is $2x$. ☐

Problem 8.17 (Derivative of a matrix map)**.** For $F(A) = A^2$ on square matrices, compute $DF(A)H$.

Solution. Expand $(A+H)^2 - A^2 = AH + HA + H^2$; the linear part is $AH + HA$ and $\|H^2\| = o(\|H\|)$. ☐

Problem 8.18 (Second-order Taylor formula)**.** Write the quadratic Taylor approximation of e^{x+y} at $(0,0)$.

Solution. Since $e^{x+y} = 1 + (x+y) + \frac{1}{2}(x+y)^2 + o(\|(x,y)\|^2)$, the quadratic approximation is $1 + x + y + \frac{1}{2}(x^2 + 2xy + y^2)$. □

Problem 8.19 (Differentiability synthesis)**.** Why are partial derivatives not the definition of differentiability?

Solution. They test only coordinate directions. Differentiability demands one linear map approximate all small directions simultaneously with an error negligible relative to distance. □

8.12 Exercises with complete solutions

Exercise 8.20. Compute the gradient of $x^2y + y^3$.

Hint. Differentiate coordinatewise. □

Solution. It is $(2xy, x^2 + 3y^2)$. □

Exercise 8.21. Find the Jacobian of $(x + y, xy)$.

Hint. Differentiate each component. □

Solution. $\begin{pmatrix} 1 & 1 \\ y & x \end{pmatrix}$. □

Exercise 8.22. Show linear maps are differentiable.

Hint. Try the map itself as derivative. □

Solution. For L, $L(a + h) - L(a) - Lh = 0$. □

Exercise 8.23. Find the Hessian of $x^2 + xy + y^2$.

Hint. Differentiate twice. □

Solution. $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. □

Exercise 8.24. What is $D(f + g)$?

Hint. Use linearity of the approximation. □

Solution. $Df + Dg$. □

Exercise 8.25. State the critical-point necessary condition.

Hint. Assume an interior extremum. □

Solution. The gradient must vanish. □

Exercise 8.26. If $Df(a) = 0$, is a necessarily an extremum?

Hint. Think of a cubic. □

Solution. No; $f(x) = x^3$ at 0 is a counterexample. □

Exercise 8.27. Compute $D(x \mapsto Ax)$.

Hint. It is linear already. $\square$

Solution. The derivative is the constant map A. $\square$

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 9

Inverse and Implicit Function Principles

The inverse and implicit function principles are local nonlinear analogues of solving an invertible linear system. This chapter is a fresh canonical construction because the source atlas contains no mapped legacy chapter for II/09.

Learning goals

The reader should be able to use the central definitions, prove the structural results, diagnose the role of each hypothesis, and solve representative analytic problems.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

9.1 Local invertibility

A nonlinear map with an invertible derivative should behave like its linearization on a sufficiently small neighborhood.

9.2 Inverse function theorem

Invertibility of $Df(a)$ gives a local differentiable inverse near a.

9.3 Derivative of the inverse

The inverse derivative is the inverse linear map: $D(f^{-1})(f(a)) = Df(a)^{-1}$.

9.4 Implicit equations

A system $F(x, y) = 0$ can be solved locally for y as a function of x when the y-Jacobian is invertible.

9.5 Implicit differentiation

The derivative of the implicit function is obtained by solving the differentiated linear system.

9.6 Regular level sets

Full-rank derivatives make level sets locally look like Euclidean subspaces.

9.7 Contraction proof strategy

One proof route reformulates the nonlinear equation as a contraction on a small ball.

9.8 Failure modes

Singular derivative, noninjectivity, or rank loss can destroy local solvability or uniqueness.

9.9 Core structural results

Theorem 9.1 (Inverse function theorem). *If $f : \mathbb{R}^n \to \mathbb{R}^n$ is C^1 near a and $Df(a)$ is invertible, then f is a C^1 diffeomorphism between neighborhoods of a and $f(a)$.*

Proof. After composing with $Df(a)^{-1}$, reduce to derivative near the identity. On a sufficiently small ball the nonlinear error has Lipschitz constant below one, so solving $f(x) = y$ becomes a contraction problem. Differentiability of the inverse follows from the linearized equation. □

Theorem 9.2 (Derivative of inverse). *Under the hypotheses of the inverse theorem, $D(f^{-1})(f(a)) = Df(a)^{-1}$.*

Proof. Differentiate $f^{-1} \circ f = \mathrm{id}$ and use the chain rule. □

Theorem 9.3 (Implicit function theorem). *If $F : \mathbb{R}^{n+m} \to \mathbb{R}^m$ is C^1, $F(a, b) = 0$, and $D_yF(a, b)$ is invertible, then near (a, b) the equation $F(x, y) = 0$ has a unique C^1 solution $y = g(x)$.*

Proof. Apply the inverse function theorem to $\Phi(x, y) = (x, F(x, y))$, whose derivative is block triangular with invertible diagonal blocks. □

9.10 Worked examples

Example 9.4 (One-dimensional inverse theorem). Explain why $f'(a) \neq 0$ yields local invertibility for a C^1 real function. Continuity keeps f' of one sign near a, so f is strictly monotone there. The inverse exists locally and differentiating $f(f^{-1}(y)) = y$ gives the inverse derivative.

Example 9.5 (A locally invertible map). For $f(x, y) = (e^x \cos y, e^x \sin y)$, determine where the derivative is invertible. The Jacobian determinant is e^{2x}, always positive, so the derivative is invertible everywhere; global injectivity fails because of 2π-periodicity in y.

Example 9.6 (Local versus global inverse). Why does the inverse function theorem not imply a global inverse for the previous map? The theorem is local. Distinct points differing by $(0, 2\pi k)$ have the same image despite nonsingular derivative everywhere.

Example 9.7 (Implicit circle graph)**.** Near $(0,1)$ solve $x^2+y^2=1$ for y and compute $y'(0)$. Take $F=x^2+y^2-1$; $F_y=2y=2$ at $(0,1)$, so $y=\sqrt{1-x^2}$ locally and $y'=-x/y$, hence $y'(0)=0$.

9.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 9.8 (One-dimensional inverse theorem)**.** Explain why $f'(a)\neq 0$ yields local invertibility for a C^1 real function.

Solution. Continuity keeps f' of one sign near a, so f is strictly monotone there. The inverse exists locally and differentiating $f(f^{-1}(y))=y$ gives the inverse derivative. □

Problem 9.9 (A locally invertible map)**.** For $f(x,y)=(e^x\cos y, e^x\sin y)$, determine where the derivative is invertible.

Solution. The Jacobian determinant is e^{2x}, always positive, so the derivative is invertible everywhere; global injectivity fails because of 2π-periodicity in y. □

Problem 9.10 (Local versus global inverse)**.** Why does the inverse function theorem not imply a global inverse for the previous map?

Solution. The theorem is local. Distinct points differing by $(0,2\pi k)$ have the same image despite nonsingular derivative everywhere. □

Problem 9.11 (Implicit circle graph)**.** Near $(0,1)$ solve $x^2+y^2=1$ for y and compute $y'(0)$.

Solution. Take $F=x^2+y^2-1$; $F_y=2y=2$ at $(0,1)$, so $y=\sqrt{1-x^2}$ locally and $y'=-x/y$, hence $y'(0)=0$. □

Problem 9.12 (Implicit derivative formula)**.** Derive $Dg(a)=-D_yF(a,b)^{-1}D_xF(a,b)$.

Solution. Differentiate $F(x,g(x))=0$: $D_xF+D_yF\,Dg=0$, then solve for Dg. □

Problem 9.13 (Regular level curve)**.** Show $x^2+y^2=1$ is locally a smooth curve at every point.

Solution. The gradient $(2x,2y)$ never vanishes on the unit circle, so one partial derivative is nonzero and the implicit theorem applies. □

Problem 9.14 (A singular failure)**.** What happens to $F(x,y)=y^2-x^3=0$ at the origin?

Solution. $D_yF(0,0)=0$, so the theorem does not apply. The cusp cannot be represented as a single smooth function $y=g(x)$ through the origin. □

Problem 9.15 (Coordinate straightening)**.** Explain geometrically what the implicit theorem says about a regular level set.

Solution. After a local C^1 change of coordinates, the equation becomes the coordinate condition $y=0$, so the level set is locally flat. □

Problem 9.16 (Newton linearization link)**.** Relate inverse-theorem solvability to Newton's method.

Solution. Both use invertibility of the derivative to solve a linearized correction equation. The theorem supplies local uniqueness and stability; Newton iterates attempt to realize that correction computationally. ☐

Problem 9.17 (Parameter dependence)**.** If $F(x, y) = y^3 + y - x$, show y is a smooth local function of x everywhere on the solution set.

Solution. $F_y = 3y^2 + 1 > 0$ everywhere, so the implicit theorem applies at every solution point. ☐

Problem 9.18 (Rank perspective)**.** Why is the determinant condition coordinate-dependent notation for an invariant statement?

Solution. Nonzero determinant means the derivative linear map is an isomorphism. Under basis changes the determinant is multiplied by nonzero factors, so invertibility itself is invariant. ☐

Problem 9.19 (IFT synthesis)**.** Summarize inverse and implicit theorems as nonlinear linear algebra.

Solution. An invertible derivative lets one solve all variables locally; an invertible derivative block lets one solve selected variables locally. Both are perturbative versions of solving nonsingular linear systems. ☐

9.12 Exercises with complete solutions

Exercise 9.20. Can $x \mapsto x^3$ use the inverse theorem at 0?

Hint. Check the derivative. ☐

Solution. No; $f'(0) = 0$, although a continuous inverse exists globally. ☐

Exercise 9.21. Find the inverse derivative formula in one dimension.

Hint. Differentiate an identity. ☐

Solution. $(f^{-1})'(f(a)) = 1/f'(a)$. ☐

Exercise 9.22. For $F(x, y) = x + y + xy$, compute $F_y(0, 0)$.

Hint. Differentiate in y. ☐

Solution. It is 1, so y is locally a function of x. ☐

Exercise 9.23. Solve the previous equation explicitly.

Hint. Factor in y. ☐

Solution. $y = -x/(1 + x)$ near 0. ☐

Exercise 9.24. Why does nonsingular Jacobian imply local openness?

Hint. Use local diffeomorphism. ☐

Solution. A local diffeomorphism maps neighborhoods to neighborhoods. ☐

Exercise 9.25. State a failure of global injectivity with nonsingular derivative.

Hint. Use periodicity. □

Solution. The exponential-polar map is a standard example. □

Exercise 9.26. What derivative block matters for solving $F(x, y) = 0$ for y?

Hint. Differentiate with respect to the solved variables. □

Solution. $D_y F$. □

Exercise 9.27. What is a regular value condition?

Hint. Think surjectivity. □

Solution. At every point of the level set, the derivative is surjective. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 10

Riemann Integration

Riemann integration quantifies accumulation by controlled finite sums. The theory rests on upper and lower sums, oscillation, and the principle that sufficiently fine partitions suppress local variation.

Learning goals

The reader should be able to use the central definitions, prove the structural results, diagnose the role of each hypothesis, and solve representative analytic problems.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

10.1 Partitions and sums

Tagged partitions produce Riemann sums; upper and lower sums eliminate dependence on tags.

10.2 Riemann integrability

A bounded function is integrable when upper and lower integrals coincide.

10.3 Darboux criterion

Integrability is equivalent to making the gap between upper and lower sums arbitrarily small.

10.4 Continuous functions

Uniform continuity on a compact interval makes continuous functions Riemann integrable.

10.5 Monotone functions

Monotonicity controls total oscillation and also guarantees integrability.

10.6 Algebra of integrable functions

Sums, scalar multiples, products, and absolute values preserve integrability under standard hypotheses.

10.7 Fundamental theorem

Integration and differentiation are inverse operations for continuous integrands.

10.8 Improper limits preview

Classical Riemann integration on compact intervals is the base case for later extension to unbounded domains or singular integrands.

10.9 Core structural results

Theorem 10.1 (Darboux criterion)**.** *A bounded function on $[a, b]$ is Riemann integrable iff for every $\varepsilon > 0$ there is a partition P with $U(f, P) - L(f, P) < \varepsilon$.*

Proof. One direction is immediate from equality of upper and lower integrals. Conversely a partition with small gap traps all finer upper and lower sums within epsilon, forcing the two integrals to coincide. □

Theorem 10.2 (Continuous integrability)**.** *Every continuous function on a compact interval is Riemann integrable.*

Proof. Uniform continuity makes oscillation on sufficiently short subintervals uniformly small; summing oscillation times interval length controls the Darboux gap. □

Theorem 10.3 (Fundamental theorem, Part I)**.** *If f is continuous on $[a, b]$ and $F(x) = \int_a^x f(t)dt$, then $F'(x) = f(x)$.*

Proof. The difference quotient is an average value of f over a shrinking interval; continuity makes that average converge to $f(x)$. □

10.10 Worked examples

Example 10.4 (Integral of a step function)**.** Compute the integral of $f = 2$ on $[0, 1)$ and $f = 5$ on $[1, 3]$. The integral is area sum $2 \cdot 1 + 5 \cdot 2 = 12$.

Example 10.5 (Darboux gap estimate)**.** If oscillation of f on every partition interval is at most η, bound $U - L$. The gap is at most $\eta \sum \Delta x_i = \eta(b - a)$.

Example 10.6 (Continuous integrability mechanism)**.** Explain where compactness enters the proof. Continuity on $[a, b]$ becomes uniform continuity, so one partition scale works simultaneously across the whole interval.

Example 10.7 (Monotone integrability)**.** Show an increasing bounded function on $[a, b]$ is integrable. For an equal partition into n pieces, the upper-lower gap telescopes to $(b - a)(f(b) - f(a))/n$.

10.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 10.8 (Integral of a step function)**.** Compute the integral of $f = 2$ on $[0, 1)$ and $f = 5$ on $[1, 3]$.

Solution. The integral is area sum $2 \cdot 1 + 5 \cdot 2 = 12$. □

Problem 10.9 (Darboux gap estimate)**.** If oscillation of f on every partition interval is at most η, bound $U - L$.

Solution. The gap is at most $\eta \sum \Delta x_i = \eta(b - a)$. □

Problem 10.10 (Continuous integrability mechanism)**.** Explain where compactness enters the proof.

Solution. Continuity on $[a, b]$ becomes uniform continuity, so one partition scale works simultaneously across the whole interval. □

Problem 10.11 (Monotone integrability)**.** Show an increasing bounded function on $[a, b]$ is integrable.

Solution. For an equal partition into n pieces, the upper-lower gap telescopes to $(b-a)(f(b)-f(a))/n$. □

Problem 10.12 (Integral of x)**.** Compute $\int_0^1 x\,dx$ from equal Riemann sums.

Solution. Right-endpoint sums are $n^{-2} \sum_{k=1}^{n} k = (n + 1)/(2n) \to 1/2$. □

Problem 10.13 (A single discontinuity)**.** Why is a bounded function with one jump discontinuity still Riemann integrable?

Solution. Isolate the jump in an interval of arbitrarily small length; on the remaining compact pieces continuity or small oscillation controls the Darboux gap. □

Problem 10.14 (Dirichlet function)**.** Why is $1_{\mathbb{Q}}$ not Riemann integrable on any nontrivial interval?

Solution. Every subinterval contains rationals and irrationals, so every upper sum is the interval length and every lower sum is zero. □

Problem 10.15 (Absolute-value inequality)**.** Prove $|\int f| \le \int |f|$ for Riemann-integrable f.

Solution. Pointwise $-|f| \le f \le |f|$; monotonicity of the integral gives the estimate. □

Problem 10.16 (Mean value for integrals)**.** If f is continuous on $[a, b]$, show $\int_a^b f = f(c)(b-a)$ for some c.

Solution. The integral average lies between the minimum and maximum of f; the intermediate-value theorem gives a point attaining that average. □

Problem 10.17 (FTC computation)**.** Evaluate $\int_0^1 3x^2\,dx$.

Solution. An antiderivative is x^3, so the integral is 1. □

Problem 10.18 (Additivity)**.** Explain why $\int_a^c f + \int_c^b f = \int_a^b f$.

Solution. Combine partitions on the two subintervals or compare Riemann sums; the common endpoint has no effect on the sum. ☐

Problem 10.19 (Integration synthesis)**.** Why is Riemann integration a limit of finite-dimensional data?

Solution. Each sum uses finitely many sample values and interval lengths; integrability asserts that all sufficiently fine such finite approximations converge to one common number. ☐

10.12 Exercises with complete solutions

Exercise 10.20. Compute $\int_0^2 1\,dx$.

Hint. Area of a rectangle. ☐

Solution. It is 2. ☐

Exercise 10.21. Compute $\int_0^1 x^2\,dx$.

Hint. Use FTC. ☐

Solution. It is 1/3. ☐

Exercise 10.22. Is every continuous function on $[a, b]$ integrable?

Hint. Use compactness. ☐

Solution. Yes. ☐

Exercise 10.23. Is every bounded function integrable?

Hint. Recall Dirichlet's example. ☐

Solution. No. ☐

Exercise 10.24. State additivity over adjacent intervals.

Hint. Split at c. ☐

Solution. $\int_a^b = \int_a^c + \int_c^b$. ☐

Exercise 10.25. If $f \geq 0$, what can be said about the integral?

Hint. Use monotonicity. ☐

Solution. The integral is nonnegative. ☐

Exercise 10.26. Does changing one function value change the Riemann integral?

Hint. Think of partition sums. ☐

Solution. No, for an integrable function a finite modification does not change the integral. ☐

Exercise 10.27. What is $\int_a^a f$?

Hint. Zero interval length. ☐

Solution. It is 0. ☐

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Part III

Sequences of Functions

Chapter 11

Pointwise and Uniform Convergence

Pointwise convergence controls each input separately; uniform convergence controls all inputs at one common scale. This distinction determines which global properties survive passage to limits.

Learning goals

The reader should be able to use the central definitions, prove the structural results, diagnose the role of each hypothesis, and solve representative analytic problems.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

11.1 Pointwise convergence

For each fixed x, the numerical sequence $f_n(x)$ converges.

11.2 Uniform convergence

One index N works for every x simultaneously.

11.3 Supremum norm

Uniform convergence is exactly convergence in the sup norm when that norm is finite.

11.4 Continuity under uniform limits

Uniform limits of continuous functions are continuous.

11.5 Cauchy criterion

Uniform convergence is equivalent to a uniform Cauchy condition in complete codomains.

11.6 Examples of failure

Pointwise limits can lose continuity or exchange of limits.

11.7 Uniform approximation

Uniform error bounds quantify global approximation quality.

11.8 Compact-domain tests

Monotonicity and compactness can sometimes upgrade pointwise convergence to uniform convergence.

11.9 Core structural results

Theorem 11.1 (Uniform limit theorem)**.** *A uniform limit of continuous functions is continuous.*

Proof. Choose one approximant uniformly close to the limit and use its continuity to bridge values of the limit at nearby points. ☐

Theorem 11.2 (Uniform Cauchy criterion)**.** *For complete codomain Y, a sequence $f_n : X \to Y$ converges uniformly iff for every ε the tail is uniformly epsilon-Cauchy.*

Proof. Uniform convergence implies uniform Cauchy by the triangle inequality. Conversely each pointwise tail is Cauchy and converges; the same uniform tail estimate survives in the limit. ☐

Theorem 11.3 (Dini-type principle)**.** *If continuous f_n on compact K decrease pointwise to continuous f, then convergence is uniform.*

Proof. The open sets where $f_n - f < \varepsilon$ increase and cover compact K; a finite subcover reduces to one sufficiently large index. ☐

11.10 Worked examples

Example 11.4 (Powers on $[0, 1]$)**.** Analyze $f_n(x) = x^n$ pointwise and uniformly. Pointwise the limit is 0 on $[0, 1)$ and 1 at 1, discontinuous. Uniform convergence fails because $\sup |f_n - f| = 1$.

Example 11.5 (Powers on $[0, a]$)**.** For $0 < a < 1$, show $x^n \to 0$ uniformly. $\sup_{[0,a]} x^n = a^n \to 0$.

Example 11.6 (Uniform implies pointwise)**.** Prove the implication directly. A uniform epsilon index works in particular at any fixed point.

Example 11.7 (Continuity of the uniform limit)**.** Give the three-term epsilon estimate. $|f(x) - f(a)| \leq |f - f_N|(x) + |f_N(x) - f_N(a)| + |f_N - f|(a)$, and choose the three terms below thirds of epsilon.

11.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 11.8 (Powers on $[0,1]$)**.** Analyze $f_n(x) = x^n$ pointwise and uniformly.

Solution. Pointwise the limit is 0 on $[0,1)$ and 1 at 1, discontinuous. Uniform convergence fails because $\sup|f_n - f| = 1$. ☐

Problem 11.9 (Powers on $[0,a]$)**.** For $0 < a < 1$, show $x^n \to 0$ uniformly.

Solution. $\sup_{[0,a]} x^n = a^n \to 0$. ☐

Problem 11.10 (Uniform implies pointwise)**.** Prove the implication directly.

Solution. A uniform epsilon index works in particular at any fixed point. ☐

Problem 11.11 (Continuity of the uniform limit)**.** Give the three-term epsilon estimate.

Solution. $|f(x) - f(a)| \le |f - f_N|(x) + |f_N(x) - f_N(a)| + |f_N - f|(a)$, and choose the three terms below thirds of epsilon. ☐

Problem 11.12 (Sup-norm reformulation)**.** Show $f_n \to f$ uniformly iff $\|f_n - f\|_\infty \to 0$.

Solution. This is just the definition of supremum norm and uniform epsilon control. ☐

Problem 11.13 (Nonuniform pointwise convergence)**.** Analyze $f_n(x) = x/(1 + nx)$ on $[0,1]$.

Solution. For each fixed x, the limit is 0. The maximum occurs near the endpoint and is $1/(n+1)$, so in fact convergence is uniform; this example illustrates the need to compute the supremum rather than guess. ☐

Problem 11.14 (A genuine nonuniform sequence)**.** Analyze $f_n(x) = nx(1 - x^2)^n$ on $[0,1]$.

Solution. For fixed $x > 0$ the exponential factor wins and the limit is zero, also at 0. But maxima remain of order $\sqrt{n}$, so convergence is not uniform. ☐

Problem 11.15 (Uniform Cauchy estimate)**.** If $\|f_{n+1} - f_n\|_\infty \le 2^{-n}$, prove uniform convergence in $C(K)$.

Solution. The tail is bounded by a geometric series, hence uniformly Cauchy; completeness of the sup-norm space gives uniform convergence. ☐

Problem 11.16 (Uniform approximation preserves boundedness)**.** If f_n are uniformly close to bounded f_N, show the limit is bounded.

Solution. For one large N, $|f| \le |f - f_N| + |f_N|$, and the uniform error is globally bounded. ☐

Problem 11.17 (Interchanging an outer limit)**.** If $x_m \to x$ and $f_n \to f$ uniformly with continuous f_n, explain why $f(x_m) \to f(x)$.

Solution. Uniform convergence gives continuity of f, then ordinary continuity handles the sequence x_m. ☐

Problem 11.18 (Dini mechanism)**.** Why is compactness decisive in Dini's theorem?

Solution. Pointwise convergence gives one local index per point; compactness extracts finitely many local neighborhoods and hence one maximum index working globally. ☐

Problem 11.19 (Convergence synthesis)**.** Why is uniform convergence the natural topology for preserving continuity?

Solution. Continuity is a local statement needing one approximation to work at two nearby points simultaneously. Uniform error control supplies that common approximant everywhere. □

11.12 Exercises with complete solutions

Exercise 11.20. Does uniform convergence imply pointwise?

Hint. Use definitions. □

Solution. Yes. □

Exercise 11.21. Does pointwise imply uniform?

Hint. Use x^n on $[0, 1]$. □

Solution. No. □

Exercise 11.22. If $\|f_n - f\|_\infty \leq 1/n$, what follows?

Hint. Take the bound to zero. □

Solution. Uniform convergence. □

Exercise 11.23. Can a uniform limit of continuous functions be discontinuous?

Hint. Use the uniform limit theorem. □

Solution. No. □

Exercise 11.24. Find the pointwise limit of x/n.

Hint. Fix x. □

Solution. It is 0. □

Exercise 11.25. Is $x/n \to 0$ uniform on $[0, 1]$?

Hint. Take the supremum. □

Solution. Yes. □

Exercise 11.26. Is $x/n \to 0$ uniform on $\mathbb{R}$?

Hint. Supremum is infinite. □

Solution. No. □

Exercise 11.27. State the uniform Cauchy condition.

Hint. One index for all points and tail pairs. □

Solution. For every epsilon, $d(f_m(x), f_n(x)) < \varepsilon$ for all x and all large m, n. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 12

Interchanging Limits, Derivatives and Integrals

Passing a limit through a derivative or integral requires more than pointwise convergence. The correct hypotheses combine uniform control with convergence of derivative data or domination of integral errors.

Learning goals

The reader should be able to use the central definitions, prove the structural results, diagnose the role of each hypothesis, and solve representative analytic problems.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

12.1 Limits and integrals

Uniform convergence allows Riemann integration to commute with limits on compact intervals.

12.2 Integral error estimates

The difference of integrals is controlled by interval length times the sup-norm error.

12.3 Limits and derivatives

Uniform convergence of derivatives plus convergence at one base point yields a differentiable limit.

12.4 Why derivative exchange is harder

Differentiation amplifies small-scale oscillation and therefore needs stronger hypotheses.

12.5 Series versions

The same principles apply to term-by-term integration and differentiation of function series.

12.6 Uniform convergence on compacta

Local uniform convergence is often the right compromise on noncompact domains.

12.7 Counterexamples

Pointwise convergence alone can fail spectacularly for derivatives and integrals.

12.8 Stability viewpoint

Interchange theorems are continuity statements for integral or derivative operators in suitable function-space topologies.

12.9 Core structural results

Theorem 12.1 (Uniform limit under the integral)**.** *If $f_n \to f$ uniformly on $[a, b]$ and each f_n is Riemann integrable, then f is integrable and $\int f_n \to \int f$.*

Proof. The integral difference is at most $(b-a)\|f_n - f\|_\infty$, which tends to zero; integrability follows by approximating f uniformly with an integrable f_n. ☐

Theorem 12.2 (Derivative limit theorem)**.** *If $f_n \in C^1[a, b]$, $f_n(x_0)$ converges at one point, and f_n' converges uniformly to g, then f_n converges uniformly to a differentiable f with $f' = g$.*

Proof. Write $f_n(x) = f_n(x_0) + \int_{x_0}^{x} f_n'$. Uniform convergence of derivatives gives uniform Cauchy control of the integrals, and the fundamental theorem identifies the derivative of the limit. ☐

Theorem 12.3 (Termwise integration)**.** *If $\sum f_n$ converges uniformly and each f_n is integrable, then the sum is integrable and $\int \sum f_n = \sum \int f_n$.*

Proof. Apply the uniform-limit integral theorem to the partial sums. ☐

12.10 Worked examples

Example 12.4 (Integral interchange estimate)**.** Prove $|\int_a^b f_n - \int_a^b f| \le (b-a)\|f_n - f\|_\infty$. Apply $|\int g| \le \int |g|$ and bound the integrand by its supremum.

Example 12.5 (A pointwise failure for integrals)**.** Construct continuous spikes f_n on $[0, 1]$ converging pointwise to 0 but with integrals not tending to 0. Use triangular spikes of height n and width $2/n$ centered near zero so area stays 1 while each fixed positive point eventually lies outside the support.

Example 12.6 (Termwise integration of a geometric series)**.** For $|x| \le r < 1$, integrate $\sum_{n\ge 0} x^n$ termwise. The series is uniformly convergent by the M-test, so $\int_0^r \sum x^n dx = \sum r^{n+1}/(n+1) = -\log(1-r)$.

Example 12.7 (Derivative exchange mechanism)**.** Why is convergence at one base point needed? Derivatives determine functions only up to additive constants. Convergence at one point fixes those constants.

12.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 12.8 (Integral interchange estimate)**.** Prove $|\int_a^b f_n - \int_a^b f| \le (b-a)\|f_n - f\|_\infty$.

Solution. Apply $|\int g| \le \int |g|$ and bound the integrand by its supremum. □

Problem 12.9 (A pointwise failure for integrals)**.** Construct continuous spikes f_n on $[0,1]$ converging pointwise to 0 but with integrals not tending to 0.

Solution. Use triangular spikes of height n and width $2/n$ centered near zero so area stays 1 while each fixed positive point eventually lies outside the support. □

Problem 12.10 (Termwise integration of a geometric series)**.** For $|x| \le r < 1$, integrate $\sum_{n\ge 0} x^n$ termwise.

Solution. The series is uniformly convergent by the M-test, so $\int_0^r \sum x^n dx = \sum r^{n+1}/(n+1) = -\log(1-r)$. □

Problem 12.11 (Derivative exchange mechanism)**.** Why is convergence at one base point needed?

Solution. Derivatives determine functions only up to additive constants. Convergence at one point fixes those constants. □

Problem 12.12 (Derivative counterexample)**.** Let $f_n(x) = \sin(nx)/n$. What happens to f_n and f_n'?

Solution. $f_n \to 0$ uniformly, but $f_n' = \cos(nx)$ does not converge pointwise at most points, showing uniform convergence of functions alone does not control derivatives. □

Problem 12.13 (Uniform derivative convergence)**.** If $\|f_n' - g\|_\infty \to 0$, estimate the derivative-integral error.

Solution. For any x, $|\int_{x_0}^x (f_n' - g)| \le |b-a|\|f_n' - g\|_\infty$. □

Problem 12.14 (Series of derivatives)**.** Suppose $\sum f_n'$ converges uniformly and $\sum f_n(x_0)$ converges. What follows?

Solution. The series $\sum f_n$ converges uniformly to a differentiable function whose derivative is $\sum f_n'$. □

Problem 12.15 (Local uniform convergence)**.** Why is uniform convergence on every compact interval enough for local integral interchange?

Solution. Any fixed finite integration interval is compact, so the uniform theorem applies there. □

Problem 12.16 (Continuity of integration operator)**.** View $I(f) = \int_a^b f$. What is its operator norm on $C[a,b]$ with sup norm?

Solution. $|I(f)| \le (b-a)\|f\|_\infty$, with equality for the constant function 1, so the norm is $b-a$. □

Problem 12.17 (Differentiation is not bounded in sup norm)**.** Use $f_n(x) = \sin(nx)/n$ to explain.

Solution. $\|f_n\|_\infty = 1/n \to 0$ while $\|f_n'\|_\infty = 1$, so no fixed bound $\|f'\|_\infty \le C\|f\|_\infty$ can hold. ☐

Problem 12.18 (M-test bridge)**.** Why does the Weierstrass M-test often precede termwise integration?

Solution. It supplies uniform convergence of the function series, exactly the hypothesis needed to exchange sum and integral. ☐

Problem 12.19 (Interchange synthesis)**.** What unifies limit/integral and limit/derivative theorems?

Solution. Both ask whether an operator is continuous under a chosen function-space topology. Integration is continuous in sup norm; differentiation requires a stronger topology controlling derivatives. ☐

12.12 Exercises with complete solutions

Exercise 12.20. If $f_n \to f$ uniformly, do integrals converge?

Hint. On a compact interval, yes. ☐

Solution. Yes, for Riemann-integrable terms. ☐

Exercise 12.21. Bound the integral error by sup error.

Hint. Use interval length. ☐

Solution. At most $(b-a)\|f_n - f\|_\infty$. ☐

Exercise 12.22. Does uniform convergence imply derivative convergence?

Hint. Use $\sin(nx)/n$. ☐

Solution. No. ☐

Exercise 12.23. Why one base point in derivative theorem?

Hint. Constants. ☐

Solution. Derivative data omit additive constants. ☐

Exercise 12.24. Can a uniformly convergent series be integrated termwise?

Hint. Under integrability. ☐

Solution. Yes. ☐

Exercise 12.25. What topology makes integration continuous?

Hint. Sup norm on a compact interval. ☐

Solution. The sup norm. ☐

Exercise 12.26. Is differentiation bounded in the sup norm?

Hint. Use high-frequency functions. ☐

Solution. No. ☐

Exercise 12.27. If derivatives converge uniformly to g, what candidate derivative has the limit?

Hint. Use the theorem. □

Solution. g. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 13

Infinite Series of Functions

Infinite series of functions are controlled most effectively by uniform comparison tests. Absolute and uniform convergence turn termwise algebra, integration, and sometimes differentiation into rigorous operations.

Learning goals

The reader should be able to use the central definitions, prove the structural results, diagnose the role of each hypothesis, and solve representative analytic problems.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

13.1 Partial sums

A function series is studied through its partial-sum sequence.

13.2 Uniform convergence

Uniform convergence of partial sums is the key global notion.

13.3 Weierstrass M-test

Uniform domination by a summable numerical series gives uniform absolute convergence.

13.4 Power series

Inside the radius of convergence, power series converge uniformly on compact subintervals.

13.5 Algebra of series

Uniformly convergent series can be added and scaled termwise.

13.6 Integration

Uniform convergence justifies termwise integration.

13.7 Differentiation

Uniform convergence of derivative series plus one-point convergence justifies termwise differentiation.

13.8 Remainder bounds

Majorant tails provide explicit uniform truncation error estimates.

13.9 Core structural results

Theorem 13.1 (Weierstrass M-test)**.** *If $|f_n(x)| \leq M_n$ for all x and $\sum M_n < \infty$, then $\sum f_n$ converges uniformly and absolutely.*

Proof. The numerical tail bound $\sum_{n>N} M_n$ controls every function tail uniformly, so the partial sums are uniformly Cauchy. □

Theorem 13.2 (Uniform termwise integration)**.** *A uniformly convergent series of integrable functions may be integrated termwise.*

Proof. Apply the uniform-limit integral theorem to partial sums. □

Theorem 13.3 (Power-series local uniformity)**.** *A power series converges uniformly on every closed subinterval strictly inside its radius of convergence.*

Proof. Choose a larger interior radius and dominate coefficients times r^n by a convergent geometric comparison from the root or ratio test. □

13.10 Worked examples

Example 13.4 (M-test with geometric majorant)**.** Show $\sum x^n/n^2$ converges uniformly on $[-1, 1]$. $|x^n/n^2| \leq 1/n^2$ and $\sum 1/n^2$ converges.

Example 13.5 (Uniform tail bound)**.** If $|f_n| \leq 2^{-n}$, bound the tail after N. The sup tail is at most $\sum_{n>N} 2^{-n} = 2^{-N}$.

Example 13.6 (Harmonic obstruction)**.** Why does the M-test fail for x^n/n on $[0, 1]$? The natural majorant $1/n$ diverges; indeed at $x = 1$ the series is harmonic.

Example 13.7 (Power-series compact convergence)**.** Why does $\sum x^n$ converge uniformly on $[-r, r]$ for $r < 1$? $|x^n| \leq r^n$ and the geometric series converges.

13.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 13.8 (M-test with geometric majorant)**.** Show $\sum x^n/n^2$ converges uniformly on $[-1, 1]$.

Solution. $|x^n/n^2| \leq 1/n^2$ and $\sum 1/n^2$ converges. ☐

Problem 13.9 (Uniform tail bound)**.** If $|f_n| \leq 2^{-n}$, bound the tail after N.

Solution. The sup tail is at most $\sum_{n>N} 2^{-n} = 2^{-N}$. ☐

Problem 13.10 (Harmonic obstruction)**.** Why does the M-test fail for x^n/n on $[0, 1]$?

Solution. The natural majorant $1/n$ diverges; indeed at $x = 1$ the series is harmonic. ☐

Problem 13.11 (Power-series compact convergence)**.** Why does $\sum x^n$ converge uniformly on $[-r, r]$ for $r < 1$?

Solution. $|x^n| \leq r^n$ and the geometric series converges. ☐

Problem 13.12 (Termwise integral)**.** Integrate $\sum_{n\geq 0} x^n$ on $[0, r]$, $r < 1$.

Solution. Uniform convergence gives $\sum r^{n+1}/(n + 1) = -\log(1 - r)$. ☐

Problem 13.13 (A derivative series)**.** For $\sum x^n/n!$, justify termwise differentiation on bounded intervals.

Solution. Derivative terms are $x^{n-1}/(n-1)!$, uniformly dominated on $[-R, R]$ by $R^{n-1}/(n-1)!$. ☐

Problem 13.14 (Absolute versus uniform)**.** Does pointwise absolute convergence imply uniform convergence?

Solution. No. Uniformity requires tail control independent of x; absolute convergence can deteriorate as x varies. ☐

Problem 13.15 (Uniform Cauchy criterion)**.** State it for a series.

Solution. For every epsilon there is N such that every finite tail $\sum_{n=p}^{q} f_n(x)$ has magnitude below epsilon for all x whenever $q \geq p \geq N$. ☐

Problem 13.16 (Remainder estimate)**.** If $|f_n| \leq M_n$, how can one certify truncation accuracy?

Solution. Bound the sup error by the numerical tail $\sum_{n>N} M_n$. ☐

Problem 13.17 (Uniform convergence preserves continuity)**.** Why is a uniformly convergent series of continuous functions continuous?

Solution. Its partial sums are continuous and converge uniformly, so the uniform-limit theorem applies. ☐

Problem 13.18 (Local uniformity on noncompact domains)**.** What does it mean for a series to converge locally uniformly?

Solution. It converges uniformly on every compact subset of the domain. ☐

Problem 13.19 (Series synthesis)**.** Why are majorants central?

Solution. They replace a difficult family of function tails by one numerical tail independent of the point, converting pointwise estimates into uniform convergence. ☐

13.12 Exercises with complete solutions

Exercise 13.20. Apply M-test to $\sum x^n/2^n$ on $[-1, 1]$.

Hint. Use $(1/2)^n$. ☐

Solution. It converges uniformly. ☐

Exercise 13.21. Does $\sum x^n$ converge uniformly on $[0, 1]$?

Hint. Check $x = 1$. ☐

Solution. No; it does not even converge there. ☐

Exercise 13.22. Uniform convergence of $\sum f_n$ means what for partial sums?

Hint. Use the definition. ☐

Solution. Partial sums converge uniformly. ☐

Exercise 13.23. If $\sum M_n$ converges, what about its tails?

Hint. Cauchy criterion. ☐

Solution. They tend to zero. ☐

Exercise 13.24. Can a uniformly convergent series of continuous functions be discontinuous?

Hint. No. ☐

Solution. Its sum is continuous. ☐

Exercise 13.25. What is a power-series radius?

Hint. Boundary between convergence and divergence. ☐

Solution. A number R such that convergence holds for $|x - a| < R$ and divergence for $|x - a| > R$. ☐

Exercise 13.26. Can the M-test prove absolute convergence?

Hint. Yes. ☐

Solution. It gives uniform absolute convergence. ☐

Exercise 13.27. How do you estimate a remainder under M-test?

Hint. Use majorant tail. ☐

Solution. By $\sum_{n>N} M_n$. ☐

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 14

Trigonometric Series

Trigonometric series represent periodic functions in orthogonal sine and cosine modes. The first layer of Fourier analysis concerns coefficients, orthogonality, partial sums, and the distinction between formal expansions and genuine convergence.

Learning goals

The reader should be able to use the central definitions, prove the structural results, diagnose the role of each hypothesis, and solve representative analytic problems.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

14.1 Periodic functions

Periodicity identifies functions on the line with functions on a circle.

14.2 Trigonometric system

Sines and cosines form an orthogonal family on a full period.

14.3 Fourier coefficients

Coefficients are obtained by inner products with the trigonometric modes.

14.4 Partial sums

Finite Fourier sums are trigonometric polynomials.

14.5 Bessel inequality

Orthogonality bounds the square-sum of coefficients by the L^2 energy.

14.6 Dirichlet kernel

Partial sums can be written as convolution with the Dirichlet kernel.

14.7 Pointwise convergence ideas

Smoothness and one-sided limits influence pointwise convergence.

14.8 Parseval preview

When the system is complete, coefficient energy equals function energy.

14.9 Core structural results

Theorem 14.1 (Orthogonality relations). *Distinct sine/cosine modes are orthogonal on* $[-\pi, \pi]$.

Proof. Use product-to-sum identities; integrals of nonconstant integer-frequency sines and cosines over a full period vanish. □

Theorem 14.2 (Bessel inequality). *For an orthonormal family, the sum of squared Fourier coefficients is bounded by the squared norm of the function.*

Proof. Subtract the finite orthogonal projection and expand its nonnegative squared norm. □

Theorem 14.3 (Dirichlet midpoint principle). *Under standard piecewise smooth hypotheses, Fourier sums converge at a jump to the midpoint of one-sided limits.*

Proof. Rewrite the partial sum using the Dirichlet kernel and use oscillatory cancellation away from the origin plus local one-sided regularity. □

14.10 Worked examples

Example 14.4 (Coefficient of a constant). Find the Fourier coefficients of $f(x) = 1$. Only the constant coefficient is nonzero; all sine and positive cosine modes integrate to zero.

Example 14.5 (Odd symmetry). What coefficients vanish for an odd function? The constant and cosine coefficients vanish; only sine coefficients may remain.

Example 14.6 (Even symmetry). What coefficients vanish for an even function? All sine coefficients vanish.

Example 14.7 (Orthogonality computation). Compute $\int_{-\pi}^{\pi} \cos(nx) \cos(mx) dx$ for $n \neq m$. Product-to-sum reduces it to integrals of nonconstant cosine modes, hence zero.

14.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 14.8 (Coefficient of a constant). Find the Fourier coefficients of $f(x) = 1$.

Solution. Only the constant coefficient is nonzero; all sine and positive cosine modes integrate to zero. □

Problem 14.9 (Odd symmetry)**.** What coefficients vanish for an odd function?

Solution. The constant and cosine coefficients vanish; only sine coefficients may remain. □

Problem 14.10 (Even symmetry)**.** What coefficients vanish for an even function?

Solution. All sine coefficients vanish. □

Problem 14.11 (Orthogonality computation)**.** Compute $\int_{-\pi}^{\pi} \cos(nx)\cos(mx)dx$ for $n \neq m$.

Solution. Product-to-sum reduces it to integrals of nonconstant cosine modes, hence zero. □

Problem 14.12 (Sawtooth structure)**.** Why does an odd sawtooth have a sine-only series?

Solution. Odd symmetry kills the even cosine modes and constant term. □

Problem 14.13 (Finite projection)**.** Why is the Nth Fourier partial sum the best L^2 approximation among trigonometric polynomials of degree at most N?

Solution. It is the orthogonal projection onto that finite-dimensional subspace. □

Problem 14.14 (Bessel estimate)**.** If $\|f\|_2 = 1$, what can be said about Fourier coefficients?

Solution. Their squared magnitudes have total finite partial sums bounded by 1. □

Problem 14.15 (Dirichlet kernel integral)**.** Why must the Dirichlet kernel have total mass 2π under the standard normalization?

Solution. The partial-sum operator reproduces constants, so integrating the kernel against the constant function must return that constant. □

Problem 14.16 (Jump value)**.** What Fourier value is expected at a jump from L_- to L_+?

Solution. The midpoint $(L_- + L_+)/2$ under Dirichlet hypotheses. □

Problem 14.17 (Gibbs warning)**.** What does Gibbs phenomenon say qualitatively?

Solution. Near jumps, partial sums exhibit persistent overshoot whose width shrinks but relative amplitude does not vanish. □

Problem 14.18 (Parseval meaning)**.** Interpret Parseval's identity.

Solution. It says the orthogonal mode coefficients preserve total quadratic energy. □

Problem 14.19 (Trigonometric synthesis)**.** Why are Fourier coefficients canonical?

Solution. Orthogonality isolates each mode by an inner product, making coefficients invariantly determined by projection rather than by solving a coupled system. □

14.12 Exercises with complete solutions

Exercise 14.20. What modes survive for an even function?

Hint. Parity. □

Solution. Constant and cosines. □

Exercise 14.21. What modes survive for an odd function?

Hint. Parity. □

Solution. Sines. □

Exercise 14.22. Are $\sin nx$ and $\cos mx$ orthogonal?

Hint. Parity/product identities. □

Solution. Yes on a full symmetric period. □

Exercise 14.23. What is a trigonometric polynomial?

Hint. Finite linear combination. □

Solution. A finite combination of constant, sine, and cosine modes. □

Exercise 14.24. What is Bessel inequality about?

Hint. Coefficient energy. □

Solution. It bounds coefficient square-sums by function energy. □

Exercise 14.25. At a smooth point, what value is expected from Fourier convergence?

Hint. The function value. □

Solution. The ordinary value. □

Exercise 14.26. At a jump, what value is expected?

Hint. Average of one-sided limits. □

Solution. Their midpoint. □

Exercise 14.27. What operator is the Fourier partial sum?

Hint. Orthogonal projection in L^2. □

Solution. Projection onto the first modes. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 15

Pathological and Nowhere-Differentiable Functions

Pathological functions reveal the sharp limits of naive intuition. Continuity need not imply differentiability, pointwise limits can destroy regularity, and dense oscillation can coexist with precise global structure.

Learning goals

The reader should be able to use the central definitions, prove the structural results, diagnose the role of each hypothesis, and solve representative analytic problems.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

15.1 Everywhere continuous, nowhere differentiable

Weierstrass-type series show continuity can coexist with complete failure of classical tangents.

15.2 Oscillation across scales

High-frequency components with slowly decaying amplitude create roughness at every scale.

15.3 Cantor-type functions

Monotone continuous functions can have derivative zero on a large set yet still increase globally.

15.4 Pointwise-limit pathologies

Limits can lose continuity when convergence is not uniform.

15.5 Dense discontinuities

Functions may be discontinuous on dense sets while retaining integrability properties.

15.6 Baire viewpoint

Category methods explain why rough functions are abundant rather than exceptional.

15.7 Moduli of continuity

Hölder estimates quantify regularity below differentiability.

15.8 Lessons for hypotheses

Counterexamples identify which assumptions in major theorems are genuinely necessary.

15.9 Core structural results

Theorem 15.1 (Uniform convergence of a Weierstrass series)**.** *If* $\sum a^n \cos(b^n x)$ *has* $0 < a < 1$*, then it converges uniformly and defines a continuous function.*

Proof. Each term is bounded by a^n and the geometric majorant converges; the uniform limit of continuous functions is continuous. □

Theorem 15.2 (Cantor-function monotonicity)**.** *The Cantor function is continuous, nondecreasing, and constant on every removed middle-third interval.*

Proof. Its ternary/binary digit construction preserves order; uniform approximants are continuous monotone functions and stabilize on removed intervals. □

Theorem 15.3 (Hölder roughness principle)**.** *A Hölder estimate* $|f(x) - f(y)| \leq C|x - y|^\alpha$ *with* $0 < \alpha < 1$ *permits much rougher behavior than Lipschitz regularity.*

Proof. The quotient by $|x - y|$ may grow like $|x - y|^{\alpha-1}$, so differentiability is not forced. □

15.10 Worked examples

Example 15.4 (Uniform continuity of Weierstrass partial sums)**.** Why does the geometric amplitude guarantee a continuous limit? The M-test gives uniform convergence, and uniform limits preserve continuity.

Example 15.5 (Derivative heuristic)**.** Why can differentiating $a^n \cos(b^n x)$ termwise be dangerous when $ab > 1$? Derivative amplitudes behave like $(ab)^n$, which grow rather than decay, signaling loss of derivative control.

Example 15.6 (Cantor function and derivative zero)**.** How can a nonconstant monotone function have derivative zero on a dense open set? It is constant on each complementary middle-third interval; the union of these intervals is dense and open, yet the function increases through the Cantor set.

Example 15.7 (Pointwise discontinuous limit)**.** Use x^n on $[0, 1]$ to exhibit a discontinuous pointwise limit. The limit is 0 on $[0, 1)$ and 1 at 1.

15.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 15.8 (Uniform continuity of Weierstrass partial sums)**.** Why does the geometric amplitude guarantee a continuous limit?

Solution. The M-test gives uniform convergence, and uniform limits preserve continuity. ☐

Problem 15.9 (Derivative heuristic)**.** Why can differentiating $a^n \cos(b^n x)$ termwise be dangerous when $ab > 1$?

Solution. Derivative amplitudes behave like $(ab)^n$, which grow rather than decay, signaling loss of derivative control. ☐

Problem 15.10 (Cantor function and derivative zero)**.** How can a nonconstant monotone function have derivative zero on a dense open set?

Solution. It is constant on each complementary middle-third interval; the union of these intervals is dense and open, yet the function increases through the Cantor set. ☐

Problem 15.11 (Pointwise discontinuous limit)**.** Use x^n on $[0, 1]$ to exhibit a discontinuous pointwise limit.

Solution. The limit is 0 on $[0, 1)$ and 1 at 1. ☐

Problem 15.12 (Dirichlet function)**.** What is special about $1_{\mathbb{Q}}$?

Solution. It is discontinuous at every real point because every interval contains both rationals and irrationals. ☐

Problem 15.13 (Thomae contrast)**.** Describe why Thomae's function is discontinuous at rationals but continuous at irrationals.

Solution. Large values occur only at rationals with small denominators, of which there are finitely many in a bounded interval; near an irrational one can avoid those finitely many. ☐

Problem 15.14 (Hölder but not Lipschitz)**.** Show $f(x) = \sqrt{|x|}$ is Hölder-1/2 but not Lipschitz near zero.

Solution. $|\sqrt{|x|} - \sqrt{|y|}| \leq \sqrt{|x - y|}$, while $|f(x) - f(0)|/|x| = 1/\sqrt{|x|} \to \infty$. ☐

Problem 15.15 (Dense set of discontinuities)**.** Can a Riemann-integrable function have a dense set of discontinuities?

Solution. Yes; Thomae's function is Riemann integrable and discontinuous at every rational, a dense set of measure zero. ☐

Problem 15.16 (Nowhere differentiability lesson)**.** Why does continuity alone not control local slope?

Solution. Continuity bounds function increments absolutely, but differentiability requires increments to be asymptotically linear relative to distance. ☐

Problem 15.17 (Uniform-limit protection)**.** Which pathology does uniform convergence prevent?

Solution. It prevents continuous approximants from converging to a discontinuous limit. □

Problem 15.18 (Hypothesis testing)**.** How should a counterexample be used when reading a theorem?

Solution. Remove one hypothesis at a time and see whether the conclusion fails; a sharp counterexample shows that hypothesis controls a real mechanism. □

Problem 15.19 (Pathology synthesis)**.** Why are pathological examples central to analysis?

Solution. They separate intuition based on smooth finite-dimensional examples from the exact logical content of definitions and theorems. □

15.12 Exercises with complete solutions

Exercise 15.20. Is every continuous function differentiable somewhere?

Hint. Think of nowhere-differentiable examples. □

Solution. No. □

Exercise 15.21. Does uniform limit of continuous functions stay continuous?

Hint. Yes. □

Solution. Yes. □

Exercise 15.22. Does pointwise limit?

Hint. Use x^n. □

Solution. Not necessarily. □

Exercise 15.23. Can a monotone function be constant on dense open pieces yet nonconstant globally?

Hint. Cantor function. □

Solution. Yes. □

Exercise 15.24. Is $\sqrt{|x|}$ Lipschitz near zero?

Hint. Check quotient. □

Solution. No. □

Exercise 15.25. Is it Hölder-1/2?

Hint. Use square-root inequality. □

Solution. Yes. □

Exercise 15.26. Is Dirichlet's function Riemann integrable?

Hint. Upper/lower sums. □

Solution. No. □

Exercise 15.27. Why study counterexamples?

Hint. Test hypotheses. □

Solution. They reveal necessity and boundaries of theorems. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Part IV

Fixed Points and Differential Equations

Chapter 16

Contraction Mappings

The contraction mapping theorem turns a quantitative shrinking estimate into existence, uniqueness, and an algorithm. It is one of the cleanest bridges between analysis and computation.

Learning goals

The reader should be able to use the central definitions, prove the structural results, diagnose the role of each hypothesis, and solve representative analytic problems.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

16.1 Contractions

A contraction reduces all distances by a fixed factor $q < 1$.

16.2 Fixed points

A fixed point solves $Tx = x$.

16.3 Picard iteration

Repeated application $x_{n+1} = Tx_n$ is the natural algorithm.

16.4 Completeness

The iteration is Cauchy because geometric distance estimates telescope; completeness supplies the limit.

16.5 Uniqueness

Two fixed points would be strictly closer after applying T, impossible unless equal.

16.6 Error estimates

A priori and a posteriori geometric bounds quantify iteration error.

16.7 Parameter dependence

Contractive fixed points vary stably under perturbations.

16.8 Applications

Integral equations and ODEs can be reformulated as fixed-point problems.

16.9 Core structural results

Theorem 16.1 (Banach fixed-point theorem). *A contraction on a nonempty complete metric space has a unique fixed point, and every Picard iteration converges to it.*

Proof. The successive differences decay geometrically, making iterates Cauchy. Completeness gives a limit; continuity of the contraction makes it fixed; the contraction inequality gives uniqueness. □

Theorem 16.2 (A priori error bound). *If $x_{n+1} = Tx_n$ and contraction factor is q, then $d(x_n, x_*) \leq q^n(1-q)^{-1}d(x_1, x_0)$.*

Proof. Bound the tail by the geometric sum of successive differences. □

Theorem 16.3 (Stability under perturbation). *If contractions T, S share factor $q < 1$ and fixed points x_T, x_S, then $d(x_T, x_S) \leq (1-q)^{-1} \sup_x d(Tx, Sx)$.*

Proof. Insert and subtract Tx_S and use the contraction bound before rearranging. □

16.10 Worked examples

Example 16.4 (Affine contraction). Solve $x = (x+3)/4$. The map has factor $1/4$ and fixed point $x = 1$.

Example 16.5 (Iteration rate). How many powers of q control the error? Error decays geometrically like q^n, up to the prefactor determined by the first step.

Example 16.6 (Uniqueness proof). Show two fixed points must coincide. $d(x, y) = d(Tx, Ty) \leq qd(x, y)$; since $q < 1$, the distance is zero.

Example 16.7 (Why completeness matters). Give a contraction on an incomplete space with no fixed point. On $(0, 1)$, $T(x) = (x + 1)/2$ is a contraction whose fixed point 1 lies outside the space.

16.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 16.8 (Affine contraction). Solve $x = (x+3)/4$.

Solution. The map has factor $1/4$ and fixed point $x = 1$. ☐

Problem 16.9 (Iteration rate)**.** How many powers of q control the error?

Solution. Error decays geometrically like q^n, up to the prefactor determined by the first step. ☐

Problem 16.10 (Uniqueness proof)**.** Show two fixed points must coincide.

Solution. $d(x, y) = d(Tx, Ty) \leq qd(x, y)$; since $q < 1$, the distance is zero. ☐

Problem 16.11 (Why completeness matters)**.** Give a contraction on an incomplete space with no fixed point.

Solution. On $(0, 1)$, $T(x) = (x + 1)/2$ is a contraction whose fixed point 1 lies outside the space. ☐

Problem 16.12 (Matrix contraction)**.** When is $T(x) = Ax + b$ a Euclidean contraction?

Solution. Whenever the operator norm $\|A\|_2 < 1$. ☐

Problem 16.13 (Fixed-point series)**.** Solve $x = Ax + b$ with $\|A\| < 1$.

Solution. $x = (I - A)^{-1}b = \sum_{n\geq 0} A^n b$, with convergence from the geometric operator-norm bound. ☐

Problem 16.14 (A posteriori estimate)**.** Bound $d(x_n, x_*)$ using $d(x_n, x_{n-1})$.

Solution. The tail is at most $d(x_n, x_{n-1})/(1 - q)$. ☐

Problem 16.15 (Nested mapping)**.** Why must T map the complete set into itself?

Solution. Otherwise the iteration may leave the domain and the completeness argument no longer applies. ☐

Problem 16.16 (Parameter stability)**.** What happens if T is perturbed by at most epsilon uniformly?

Solution. The fixed point moves by at most $\varepsilon/(1 - q)$. ☐

Problem 16.17 (Integral-equation preview)**.** How does an integral operator become a contraction?

Solution. Bound the sup-norm difference of outputs by an interval-length times a Lipschitz constant times the input sup difference. ☐

Problem 16.18 (ODE preview)**.** How does Picard iteration arise for $y' = f(t, y)$?

Solution. Rewrite as $y(t) = y_0 + \int f(s, y(s))ds$ and iterate the integral operator. ☐

Problem 16.19 (Contraction synthesis)**.** Why is this theorem algorithmic?

Solution. Its proof constructs the solution by iteration and simultaneously provides computable geometric error bounds. ☐

16.12 Exercises with complete solutions

Exercise 16.20. Find fixed point of $T(x) = x/2 + 1$.

Hint. Solve algebraically. □

Solution. It is 2. □

Exercise 16.21. What is the contraction factor?

Hint. Coefficient of x. □

Solution. 1/2. □

Exercise 16.22. Why unique?

Hint. Use fixed-point inequality. □

Solution. Two fixed points have zero distance. □

Exercise 16.23. Does a contraction have to be linear?

Hint. No. □

Solution. No. □

Exercise 16.24. What property supplies the limit?

Hint. Completeness. □

Solution. Completeness of the space. □

Exercise 16.25. What sequence is used?

Hint. Picard iteration. □

Solution. $x_{n+1} = Tx_n$. □

Exercise 16.26. What is the qualitative error rate?

Hint. Geometric. □

Solution. Order q^n. □

Exercise 16.27. Can $q = 1$ be used?

Hint. No strict contraction. □

Solution. Not in Banach's theorem. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 17

Brouwer-Type Fixed-Point Ideas

Brouwer-type fixed-point ideas replace metric contraction by topology. Existence can survive without uniqueness or an iterative rate, revealing a distinct mechanism from Banach's theorem.

Learning goals

The reader should be able to use the central definitions, prove the structural results, diagnose the role of each hypothesis, and solve representative analytic problems.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

17.1 Fixed points without contraction

Continuous maps can have fixed points for topological rather than metric reasons.

17.2 Interval case

The intermediate-value theorem already yields a one-dimensional Brouwer theorem.

17.3 Convex compact sets

Brouwer's theorem applies to continuous self-maps of compact convex subsets of Euclidean space.

17.4 No-retraction intuition

A ball cannot continuously retract onto its boundary.

17.5 Degree intuition

Topological degree counts preimages with orientation and obstructs disappearance of fixed points.

17.6 Existence versus uniqueness

Brouwer gives existence but generally no uniqueness.

17.7 Applications

Equilibria in games, economics, and nonlinear systems often use Brouwer-type existence.

17.8 Comparison with Banach

Banach is quantitative and metric; Brouwer is qualitative and topological.

17.9 Core structural results

Theorem 17.1 (Interval fixed-point theorem)**.** *Every continuous $f : [a, b] \to [a, b]$ has a fixed point.*

Proof. Apply the intermediate-value theorem to $g(x) = f(x) - x$: $g(a) \geq 0$ and $g(b) \leq 0$. □

Theorem 17.2 (Brouwer fixed-point theorem)**.** *Every continuous map from a nonempty compact convex subset of $\mathbb{R}^n$ to itself has a fixed point.*

Proof. The full proof uses a topological obstruction such as no-retraction, Sperner's lemma, or degree; the key point is that a fixed-point-free map would induce impossible boundary data. □

Theorem 17.3 (No-retraction consequence)**.** *There is no continuous retraction from the closed ball onto its boundary sphere.*

Proof. A retraction would contradict Brouwer by composing radial constructions to obtain a fixed-point-free self-map of the ball. □

17.10 Worked examples

Example 17.4 (Interval proof)**.** Prove the fixed-point theorem on $[0, 1]$. For $g = f - \mathrm{id}$, $g(0) \geq 0$ and $g(1) \leq 0$; continuity gives a zero.

Example 17.5 (Failure without compactness)**.** Give a continuous self-map of $(0, 1)$ with no fixed point. $f(x) = (x + 1)/2$ maps $(0, 1)$ to itself and has only fixed point 1, outside the domain.

Example 17.6 (Failure without self-map condition)**.** Why does $f(x) = x + 1$ on $[0, 1]$ not contradict Brouwer? Its image is not contained in $[0, 1]$.

Example 17.7 (Nonuniqueness)**.** Give a Brouwer map with many fixed points. The identity map fixes every point.

17.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 17.8 (Interval proof)**.** Prove the fixed-point theorem on $[0, 1]$.

Solution. For $g = f - \mathrm{id}$, $g(0) \geq 0$ and $g(1) \leq 0$; continuity gives a zero. ☐

Problem 17.9 (Failure without compactness)**.** Give a continuous self-map of $(0, 1)$ with no fixed point.

Solution. $f(x) = (x + 1)/2$ maps $(0, 1)$ to itself and has only fixed point 1, outside the domain. ☐

Problem 17.10 (Failure without self-map condition)**.** Why does $f(x) = x + 1$ on $[0, 1]$ not contradict Brouwer?

Solution. Its image is not contained in $[0, 1]$. ☐

Problem 17.11 (Nonuniqueness)**.** Give a Brouwer map with many fixed points.

Solution. The identity map fixes every point. ☐

Problem 17.12 (Banach versus Brouwer)**.** Which theorem gives an iteration rate?

Solution. Banach does; Brouwer generally supplies existence only. ☐

Problem 17.13 (Convexity role)**.** Why is an annulus not covered directly by Brouwer's convex-set statement?

Solution. An annulus is not convex; straight segments between points may leave it. ☐

Problem 17.14 (A rotation of a disk)**.** Why must a disk rotation have a fixed point?

Solution. The center is fixed, consistent with Brouwer. ☐

Problem 17.15 (Sperner intuition)**.** How does Sperner's lemma relate to Brouwer?

Solution. A labeled triangulation forces a fully labeled simplex; shrinking mesh sizes produce approximate fixed points whose compactness limit is an exact one. ☐

Problem 17.16 (Equilibrium interpretation)**.** How can a fixed point encode equilibrium?

Solution. A response map sends a state to its updated best-response or normalized state; a fixed point is unchanged by the update and thus represents equilibrium. ☐

Problem 17.17 (Topological obstruction)**.** What does 'no retraction' mean intuitively?

Solution. One cannot continuously push every interior point to the boundary while leaving boundary points fixed. ☐

Problem 17.18 (Approximate fixed points)**.** Why are compactness arguments natural in Brouwer proofs?

Solution. Combinatorial or discretized arguments produce approximate fixed points; compactness extracts a convergent subsequence and continuity turns the limiting approximation error into zero. ☐

Problem 17.19 (Brouwer synthesis)**.** What fundamentally distinguishes Brouwer from Banach?

Solution. Brouwer relies on topology of compact convex sets and gives existence; Banach relies on quantitative shrinking in complete metric spaces and gives uniqueness plus convergence. ☐

17.12 Exercises with complete solutions

Exercise 17.20. Does every continuous $[0,1] \to [0,1]$ map have a fixed point?

Hint. Yes. ☐

Solution. Yes. ☐

Exercise 17.21. Can it have several?

Hint. Identity. ☐

Solution. Yes. ☐

Exercise 17.22. Does Brouwer guarantee uniqueness?

Hint. No. ☐

Solution. No. ☐

Exercise 17.23. Does compactness matter?

Hint. Open interval counterexample. ☐

Solution. Yes. ☐

Exercise 17.24. Does convexity appear in the standard theorem?

Hint. Yes. ☐

Solution. Yes. ☐

Exercise 17.25. Is a disk convex?

Hint. Straight segments stay inside. ☐

Solution. Yes. ☐

Exercise 17.26. Is a circle convex?

Hint. Chord leaves circle. ☐

Solution. No. ☐

Exercise 17.27. What one-dimensional theorem proves the interval case?

Hint. Intermediate value theorem. ☐

Solution. IVT. ☐

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 18

Integral Equations

Integral equations encode unknown functions under an integral operator. Their analytic treatment combines norm estimates, compactness ideas, and fixed-point principles.

Learning goals

The reader should be able to use the central definitions, prove the structural results, diagnose the role of each hypothesis, and solve representative analytic problems.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

18.1 Fredholm equations

The unknown appears under an integral over a fixed domain.

18.2 Volterra equations

The upper limit depends on the evaluation point, producing a triangular time structure.

18.3 Integral operators

Kernels define linear or nonlinear operators on function spaces.

18.4 Sup-norm estimates

Kernel bounds convert integral operators into norm inequalities.

18.5 Neumann series

Small operator norm permits inversion of $I - K$ by a geometric series.

18.6 Contraction methods

Nonlinear integral equations can be solved by Banach's theorem under Lipschitz conditions.

18.7 Iterated kernels

Repeated substitution creates iterated kernel integrals.

18.8 Connection to ODEs

Initial-value ODEs are equivalent to Volterra integral equations.

18.9 Core structural results

Theorem 18.1 (Neumann-series inversion)**.** *If a bounded linear operator K satisfies $\|K\| < 1$, then $I - K$ is invertible with $(I - K)^{-1} = \sum_{n\geq 0} K^n$.*

Proof. The operator series converges absolutely in norm, and finite geometric identities pass to the limit. □

Theorem 18.2 (Kernel sup bound)**.** *For $(Kf)(x) = \int_a^b k(x,t)f(t)dt$, $\|K\|_\infty \leq (b-a)\|k\|_\infty$.*

Proof. Take absolute values, bound both kernel and function by sup norms, and integrate interval length. □

Theorem 18.3 (Volterra contraction on short intervals)**.** *If $F(t, y)$ is Lipschitz in y with constant L, the integral map $Ty = y_0 + \int_{t_0}^t F(s, y(s))ds$ is a contraction on an interval of length h when $Lh < 1$.*

Proof. The output difference is at most Lh times the input sup difference. □

18.10 Worked examples

Example 18.4 (Constant kernel)**.** Solve $f(x) = 1 + \lambda \int_0^1 f(t)dt$. Let $c = \int_0^1 f$. Then $f = 1 + \lambda c$ is constant and $c = 1 + \lambda c$, so for $\lambda \neq 1$, $f = 1/(1-\lambda)$.

Example 18.5 (Norm of a simple kernel operator)**.** For $Kf(x) = \int_0^1 xtf(t)dt$, bound $\|K\|_\infty$. Since $|xt| \leq 1$, the general bound gives $\|K\| \leq 1$; the sharper bound is $1/2$ because $\sup_x \int_0^1 xtdt = 1/2$.

Example 18.6 (Neumann approximation)**.** If $\|K\| = q < 1$, bound truncation after N terms. The inverse-series tail has norm at most $q^{N+1}/(1-q)$.

Example 18.7 (Volterra uniqueness)**.** Why are Volterra equations naturally compatible with iteration? The value at time t depends only on earlier times up to t, producing causal nested integrals and strong factorial-type bounds.

18.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 18.8 (Constant kernel)**.** Solve $f(x) = 1 + \lambda \int_0^1 f(t)dt$.

Solution. Let $c = \int_0^1 f$. Then $f = 1 + \lambda c$ is constant and $c = 1 + \lambda c$, so for $\lambda \neq 1$, $f = 1/(1-\lambda)$. □

Problem 18.9 (Norm of a simple kernel operator)**.** For $Kf(x) = \int_0^1 xtf(t)dt$, bound $\|K\|_\infty$.

Solution. Since $|xt| \le 1$, the general bound gives $\|K\| \le 1$; the sharper bound is $1/2$ because $\sup_x \int_0^1 xtdt = 1/2$. ☐

Problem 18.10 (Neumann approximation)**.** If $\|K\| = q < 1$, bound truncation after N terms.

Solution. The inverse-series tail has norm at most $q^{N+1}/(1-q)$. ☐

Problem 18.11 (Volterra uniqueness)**.** Why are Volterra equations naturally compatible with iteration?

Solution. The value at time t depends only on earlier times up to t, producing causal nested integrals and strong factorial-type bounds. ☐

Problem 18.12 (ODE equivalence)**.** Convert $y' = f(t,y)$, $y(t_0) = y_0$ to an integral equation.

Solution. Integrate to obtain $y(t) = y_0 + \int_{t_0}^t f(s, y(s))ds$. ☐

Problem 18.13 (Linear resolvent idea)**.** What does $(I - \lambda K)^{-1}$ represent?

Solution. It maps forcing terms to solutions of $f - \lambda Kf = g$ when the inverse exists. ☐

Problem 18.14 (Short-interval contraction)**.** If Lipschitz constant is $L = 3$, how small should interval length be for direct Banach contraction?

Solution. Choose $h < 1/3$. ☐

Problem 18.15 (Iterated operator estimate)**.** If $\|K\| \le q$, what is $\|K^n\|$?

Solution. At most q^n by submultiplicativity. ☐

Problem 18.16 (Kernel continuity)**.** Why does a continuous kernel on a compact square define a bounded sup-norm operator?

Solution. The kernel is bounded by compactness, and the kernel sup bound applies. ☐

Problem 18.17 (Nonlinear integral map)**.** How does a Lipschitz nonlinearity enter the estimate?

Solution. Take the difference inside the integral and use the pointwise Lipschitz bound before integration. ☐

Problem 18.18 (Fixed-point viewpoint)**.** What equation is actually solved by Banach?

Solution. The integral equation is rewritten as $f = Tf$ and the solution is the unique fixed point of T. ☐

Problem 18.19 (Integral-equation synthesis)**.** Why are integral equations often easier than differential equations?

Solution. They encode initial conditions automatically and replace derivatives by averaging operators, which are better behaved in norm estimates. ☐

18.12 Exercises with complete solutions

Exercise 18.20. What operator solves $f = g + Kf$?

Hint. Rearrange. □

Solution. $(I - K)f = g$. □

Exercise 18.21. When does Neumann series converge?

Hint. Operator norm below one. □

Solution. If $\|K\| < 1$. □

Exercise 18.22. What norm is convenient on $C[a, b]$?

Hint. Sup norm. □

Solution. $\|f\|_\infty$. □

Exercise 18.23. What does a Volterra upper limit look like?

Hint. Variable upper endpoint. □

Solution. $\int_a^x$. □

Exercise 18.24. What does a Fredholm equation use?

Hint. Fixed interval. □

Solution. $\int_a^b$. □

Exercise 18.25. How do ODEs connect?

Hint. Integrate once. □

Solution. They become Volterra equations. □

Exercise 18.26. Why does short interval help?

Hint. Multiplies Lipschitz constant. □

Solution. It makes $Lh < 1$. □

Exercise 18.27. Does $\|K\| < 1$ imply uniqueness for $(I - K)f = g$?

Hint. Yes. □

Solution. Yes, since $I - K$ is invertible. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 19

Elementary Existence Theory for ODEs

Elementary ODE existence theory is most transparent when an initial-value problem is rewritten as an integral fixed-point equation. Continuity gives existence mechanisms; Lipschitz control supplies uniqueness.

Learning goals

The reader should be able to use the central definitions, prove the structural results, diagnose the role of each hypothesis, and solve representative analytic problems.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

19.1 Initial-value problems

An ODE with initial data asks for a curve satisfying a differential rule and a point condition.

19.2 Integral formulation

Integrating converts $y' = f(t, y)$ into a Volterra equation.

19.3 Picard iteration

Successive substitution generates approximate solutions.

19.4 Local existence

Continuity and compactness can yield existence; Lipschitz hypotheses enable Banach's theorem.

19.5 Uniqueness

Lipschitz dependence on the state variable prevents branching solutions.

19.6 Continuation

Solutions extend while they remain in regions where the hypotheses stay controlled.

19.7 Continuous dependence

Small changes in initial data or vector field produce controlled changes in solutions.

19.8 Autonomous systems

Phase-space geometry reframes $y' = F(y)$ as a flow problem.

19.9 Core structural results

Theorem 19.1 (Picard–Lindelöf). *If f is continuous and locally Lipschitz in y near (t_0, y_0), then the IVP $y' = f(t, y)$, $y(t_0) = y_0$ has a unique local solution.*

Proof. On a sufficiently small rectangle, the integral operator maps a closed sup-norm ball into itself and is a contraction; Banach gives a unique fixed point. □

Theorem 19.2 (Gronwall inequality). *If $u(t) \le A + B \int_0^t u(s)ds$ with $u \ge 0$, then $u(t) \le Ae^{Bt}$.*

Proof. Define the right-hand side majorant and differentiate it; multiplying by e^{-Bt} yields a monotone quantity that bounds u. □

Theorem 19.3 (Continuous dependence). *Under a Lipschitz bound L, two solutions satisfy an exponential stability estimate controlled by initial separation.*

Proof. Subtract the integral equations, estimate by the initial difference plus L times the integral of the difference, then apply Gronwall. □

19.10 Worked examples

Example 19.4 (Linear scalar IVP). Solve $y' = ay$, $y(0) = y_0$. The solution is $y(t) = y_0 e^{at}$.

Example 19.5 (Integral reformulation). Rewrite $y' = t + y$, $y(0) = 1$. $y(t) = 1 + \int_0^t (s + y(s))ds$.

Example 19.6 (Picard first iterate). Starting from $y_0(t) = 1$, compute one Picard iterate for the previous equation. $y_1(t) = 1 + \int_0^t (s + 1)ds = 1 + t + t^2/2$.

Example 19.7 (Uniqueness failure without Lipschitz). Why can $y' = \sqrt{|y|}$, $y(0) = 0$ have nonunique solutions? The square-root field is not Lipschitz at zero; solutions may remain zero for a while and then depart along a translated quadratic branch.

19.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 19.8 (Linear scalar IVP). Solve $y' = ay$, $y(0) = y_0$.

Solution. The solution is $y(t) = y_0 e^{at}$. □

Problem 19.9 (Integral reformulation)**.** Rewrite $y' = t + y$, $y(0) = 1$.

Solution. $y(t) = 1 + \int_0^t (s + y(s))ds$. ☐

Problem 19.10 (Picard first iterate)**.** Starting from $y_0(t) = 1$, compute one Picard iterate for the previous equation.

Solution. $y_1(t) = 1 + \int_0^t (s + 1)ds = 1 + t + t^2/2$. ☐

Problem 19.11 (Uniqueness failure without Lipschitz)**.** Why can $y' = \sqrt{|y|}$, $y(0) = 0$ have nonunique solutions?

Solution. The square-root field is not Lipschitz at zero; solutions may remain zero for a while and then depart along a translated quadratic branch. ☐

Problem 19.12 (Local contraction scale)**.** If $|f(t, y) - f(t, z)| \leq L|y - z|$, what interval condition gives direct contraction?

Solution. Choose interval half-width or length h with $Lh < 1$. ☐

Problem 19.13 (Gronwall stability)**.** If two solutions have initial gap δ, bound their later gap.

Solution. Under Lipschitz constant L, the gap is at most $\delta e^{L|t-t_0|}$. ☐

Problem 19.14 (Blow-up and continuation)**.** Why does $y' = y^2$, $y(0) = 1$ not extend for all positive time?

Solution. The solution $y = 1/(1 - t)$ blows up at $t = 1$, leaving every bounded region where local continuation could proceed. ☐

Problem 19.15 (Autonomous equilibrium)**.** What is an equilibrium of $y' = F(y)$?

Solution. A point y_* with $F(y_*) = 0$, giving the constant solution $y(t) = y_*$. ☐

Problem 19.16 (Separable example)**.** Solve $y' = -2y$, $y(0) = 3$.

Solution. $y(t) = 3e^{-2t}$. ☐

Problem 19.17 (Parameter dependence)**.** Why does a Lipschitz theorem imply robustness?

Solution. The same Gronwall estimate controlling uniqueness also bounds changes caused by perturbed initial data or forcing. ☐

Problem 19.18 (Picard convergence)**.** What makes successive approximations converge?

Solution. The integral operator is a contraction on a sufficiently small complete function-space ball. ☐

Problem 19.19 (ODE synthesis)**.** Why is integration the natural foundation for existence theory?

Solution. Differential equations ask for unknown derivatives, whereas the integral equation defines a self-map on a space of continuous functions where completeness and norm estimates can be applied. ☐

19.12 Exercises with complete solutions

Exercise 19.20. Solve $y' = y$, $y(0) = 2$.

Hint. Exponential. □

Solution. $2e^t$. □

Exercise 19.21. What theorem gives local uniqueness?

Hint. Picard–Lindelöf. □

Solution. Picard–Lindelöf. □

Exercise 19.22. What hypothesis controls uniqueness?

Hint. Lipschitz in state. □

Solution. Local Lipschitz continuity in y. □

Exercise 19.23. What does Gronwall control?

Hint. Growth of differences. □

Solution. Exponential stability bounds. □

Exercise 19.24. Can solutions blow up in finite time?

Hint. Yes. □

Solution. Yes, e.g. $y' = y^2$. □

Exercise 19.25. What is Picard iteration applied to?

Hint. Integral map. □

Solution. The Volterra integral formulation. □

Exercise 19.26. What is an equilibrium?

Hint. Zero of vector field. □

Solution. A point $F(y_*) = 0$. □

Exercise 19.27. Does continuity alone always imply uniqueness?

Hint. No. □

Solution. No. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Part V

Approximation

Chapter 20

Polynomial Interpolation

Polynomial interpolation reconstructs a polynomial from finite data. Lagrange and Newton forms expose uniqueness, computation, divided differences, and the remainder term.

Learning goals

The reader should be able to use the central definitions, prove the structural results, diagnose the role of each hypothesis, and solve representative analytic problems.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

20.1 Interpolation problem

Given distinct nodes and values, seek a polynomial matching all data.

20.2 Lagrange basis

Cardinal polynomials isolate one node at a time.

20.3 Uniqueness

A polynomial of degree at most n matching $n + 1$ distinct data points is unique.

20.4 Newton form

Divided differences produce a nested incremental representation.

20.5 Divided differences

Recursive coefficients encode discrete derivatives.

20.6 Error formula

The interpolation remainder involves the next derivative and the node polynomial.

20.7 Node choice

Equally spaced nodes can be unstable at high degree; special nodes reduce worst-case error.

20.8 Hermite preview

Derivative data can be incorporated by repeated-node ideas.

20.9 Core structural results

Theorem 20.1 (Interpolation existence and uniqueness)**.** *For $n+1$ distinct nodes, there is a unique polynomial of degree at most n taking prescribed values.*

Proof. Lagrange basis polynomials give existence. The difference of two interpolants has $n+1$ roots but degree at most n, so it is zero. □

Theorem 20.2 (Lagrange formula)**.** *The interpolant is $p(x) = \sum_{j=0}^{n} y_j L_j(x)$ with $L_j(x_i) = \delta_{ij}$.*

Proof. By construction each basis polynomial is one at its own node and zero at the others, so the sum matches all data. □

Theorem 20.3 (Interpolation remainder)**.** *If $f \in C^{n+1}$, then $f(x) - p_n(x) = f^{(n+1)}(\xi)(\prod_{j=0}^{n}(x - x_j))/(n+1)!$ for a suitable ξ.*

Proof. Subtract a multiple of the node polynomial chosen to vanish also at x, then apply Rolle's theorem repeatedly to the resulting function. □

20.10 Worked examples

Example 20.4 (Two-point interpolation)**.** Interpolate $(0,1)$ and $(1,3)$. The unique linear interpolant is $p(x) = 1 + 2x$.

Example 20.5 (Three-point Lagrange)**.** Interpolate $f(0) = 0, f(1) = 1, f(2) = 4$. The data lie on x^2, so uniqueness gives $p(x) = x^2$.

Example 20.6 (Uniqueness by roots)**.** Why cannot two degree-n interpolants agree at $n+1$ distinct nodes and differ elsewhere? Their difference would be a nonzero degree-at-most-n polynomial with too many roots.

Example 20.7 (Lagrange basis property)**.** Verify $L_j(x_i) = \delta_{ij}$. At $i = j$ all factors equal one; for $i \neq j$ one numerator factor vanishes.

20.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 20.8 (Two-point interpolation)**.** Interpolate $(0,1)$ and $(1,3)$.

Solution. The unique linear interpolant is $p(x) = 1 + 2x$. □

Problem 20.9 (Three-point Lagrange)**.** Interpolate $f(0) = 0, f(1) = 1, f(2) = 4$.

Solution. The data lie on x^2, so uniqueness gives $p(x) = x^2$. □

Problem 20.10 (Uniqueness by roots)**.** Why cannot two degree-n interpolants agree at $n+1$ distinct nodes and differ elsewhere?

Solution. Their difference would be a nonzero degree-at-most-n polynomial with too many roots. □

Problem 20.11 (Lagrange basis property)**.** Verify $L_j(x_i) = \delta_{ij}$.

Solution. At $i = j$ all factors equal one; for $i \neq j$ one numerator factor vanishes. □

Problem 20.12 (Newton divided difference)**.** Compute the first divided difference for data $(x_0, y_0), (x_1, y_1)$.

Solution. It is $(y_1 - y_0)/(x_1 - x_0)$. □

Problem 20.13 (Incremental interpolation)**.** Why is Newton form convenient when adding one new node?

Solution. The previous polynomial remains intact and one new product term is appended. □

Problem 20.14 (Error node product)**.** Why do roots of the interpolation error include all nodes?

Solution. The interpolant matches the function exactly at every node. □

Problem 20.15 (Linear interpolation error)**.** State the error form for two nodes.

Solution. $f(x) - p_1(x) = f''(\xi)(x - x_0)(x - x_1)/2$. □

Problem 20.16 (Runge warning)**.** What can go wrong with high-degree equally spaced interpolation?

Solution. Endpoint oscillations can grow despite exact matching at nodes. □

Problem 20.17 (Chebyshev motivation)**.** Why choose nodes near Chebyshev points?

Solution. They reduce the maximum magnitude of the node polynomial and thereby control worst-case interpolation error. □

Problem 20.18 (Vandermonde formulation)**.** How can interpolation be written as a linear system?

Solution. Writing $p(x) = \sum a_k x^k$ gives a Vandermonde matrix whose invertibility follows from distinct nodes. □

Problem 20.19 (Interpolation synthesis)**.** What is being approximated and what is exact?

Solution. The polynomial matches finitely many data exactly, while the remainder formula quantifies error between those nodes in terms of higher derivatives and node geometry. □

20.12 Exercises with complete solutions

Exercise 20.20. Interpolate two points $(0,0),(1,1)$.

Hint. Line through them. □

Solution. $p = x$. □

Exercise 20.21. How many data points determine degree at most n?

Hint. Distinct nodes. □

Solution. $n + 1$. □

Exercise 20.22. Why distinct nodes?

Hint. Vandermonde determinant. □

Solution. Otherwise uniqueness fails or data may conflict. □

Exercise 20.23. What are Lagrange basis values at nodes?

Hint. Kronecker delta. □

Solution. 0 or 1. □

Exercise 20.24. What degree is the node polynomial?

Hint. One factor per node. □

Solution. $n + 1$. □

Exercise 20.25. What theorem proves uniqueness?

Hint. Root count. □

Solution. A nonzero degree-n polynomial has at most n roots. □

Exercise 20.26. Does interpolation guarantee small error?

Hint. No. □

Solution. Node placement and smoothness matter. □

Exercise 20.27. What form supports incremental data?

Hint. Newton form. □

Solution. Newton divided-difference form. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 21

Polynomial Approximation

Polynomial approximation asks not for exact matching at finitely many points but for small error on an entire compact interval. The Weierstrass theorem says continuous functions admit arbitrarily accurate uniform polynomial approximations.

Learning goals

The reader should be able to use the central definitions, prove the structural results, diagnose the role of each hypothesis, and solve representative analytic problems.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

21.1 Uniform approximation

Approximation quality is measured by the sup norm.

21.2 Weierstrass theorem

Polynomials are dense in $C[a, b]$.

21.3 Bernstein polynomials

On $[0, 1]$, positive averaging polynomials give a constructive proof.

21.4 Modulus of continuity

Uniform continuity controls approximation error.

21.5 Shape preservation

Positive approximation operators can preserve positivity and convexity features.

21.6 Density viewpoint

Approximation is a statement about closure of a subspace in a normed function space.

21.7 Approximation rates

Extra smoothness improves quantitative convergence rates.

21.8 Beyond polynomials

The same density philosophy leads to trigonometric and rational approximation.

21.9 Core structural results

Theorem 21.1 (Weierstrass approximation theorem)**.** *Every continuous real function on* $[a, b]$ *can be uniformly approximated by polynomials.*

Proof. After rescaling to $[0, 1]$, Bernstein polynomials converge uniformly; uniform continuity controls nearby samples and binomial concentration controls distant samples. ☐

Theorem 21.2 (Bernstein reproduction)**.** *Bernstein operators reproduce constants and the identity function.*

Proof. Binomial identities give $B_n 1 = 1$ and $B_n x = x$. ☐

Theorem 21.3 (Density consequence)**.** *For every* $f \in C[a, b]$ *and* $\varepsilon > 0$*, some polynomial* p *satisfies* $\|f - p\|_\infty < \varepsilon$*.*

Proof. This is exactly the statement that the polynomial subspace is dense in the sup-norm space. ☐

21.10 Worked examples

Example 21.4 (Approximate absolute value)**.** Why can $|x|$ on $[-1, 1]$ be uniformly approximated by polynomials despite not being differentiable at zero? Weierstrass requires continuity only, not differentiability.

Example 21.5 (Bernstein constant)**.** Show $B_n 1 = 1$. The Bernstein basis sums to $(x + (1 - x))^n = 1$.

Example 21.6 (Bernstein identity)**.** Show $B_n x = x$. The weighted sum of k/n against binomial probabilities equals the mean of a Binomial(n, x) variable divided by n, namely x.

Example 21.7 (Uniform versus pointwise approximation)**.** Why is sup norm stronger? It bounds the worst error at every point simultaneously rather than allowing the bad point to vary with the approximant.

21.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 21.8 (Approximate absolute value)**.** Why can $|x|$ on $[-1, 1]$ be uniformly approximated by polynomials despite not being differentiable at zero?

Solution. Weierstrass requires continuity only, not differentiability. ☐

Problem 21.9 (Bernstein constant)**.** Show $B_n 1 = 1$.

Solution. The Bernstein basis sums to $(x + (1 - x))^n = 1$. ☐

Problem 21.10 (Bernstein identity)**.** Show $B_n x = x$.

Solution. The weighted sum of k/n against binomial probabilities equals the mean of a Binomial(n, x) variable divided by n, namely x. ☐

Problem 21.11 (Uniform versus pointwise approximation)**.** Why is sup norm stronger?

Solution. It bounds the worst error at every point simultaneously rather than allowing the bad point to vary with the approximant. ☐

Problem 21.12 (Density interpretation)**.** What does polynomial density mean geometrically?

Solution. Every open sup-norm ball around every continuous function contains a polynomial. ☐

Problem 21.13 (No exact representation required)**.** Does Weierstrass say every continuous function is a polynomial?

Solution. No; it says polynomials can approach arbitrarily closely in sup norm. ☐

Problem 21.14 (Approximation of e^x)**.** Give one natural polynomial approximant.

Solution. Taylor polynomials provide explicit approximants; on a compact interval their remainders can be bounded uniformly. ☐

Problem 21.15 (Approximation and integration)**.** If $p_n \to f$ uniformly, what happens to integrals?

Solution. The integrals converge because integration is continuous in the sup norm. ☐

Problem 21.16 (Approximation and roots)**.** Does a uniformly close polynomial necessarily preserve every zero?

Solution. No; zeros can move or disappear unless extra sign or transversality information is available. ☐

Problem 21.17 (Smoothness and rate)**.** Why should smoother functions be easier to approximate rapidly?

Solution. Higher regularity controls oscillation and derivative growth, enabling sharper remainder and constructive approximation estimates. ☐

Problem 21.18 (Positive operators)**.** Why are Bernstein polynomials stable?

Solution. They are convex combinations of sampled values, so they do not amplify the sup norm and preserve positivity. ☐

Problem 21.19 (Approximation synthesis)**.** How does this differ from interpolation?

Solution. Interpolation enforces exact agreement at selected points; approximation minimizes or controls global error across the whole interval. ☐

21.12 Exercises with complete solutions

Exercise 21.20. What norm does Weierstrass use?

Hint. Sup norm. ☐

Solution. Uniform norm. ☐

Exercise 21.21. Must f be differentiable?

Hint. No. ☐

Solution. Continuity suffices. ☐

Exercise 21.22. What interval is standard for Bernstein polynomials?

Hint. $[0, 1]$. ☐

Solution. $[0, 1]$. ☐

Exercise 21.23. Do Bernstein polynomials reproduce constants?

Hint. Yes. ☐

Solution. Yes. ☐

Exercise 21.24. Do they reproduce x?

Hint. Yes. ☐

Solution. Yes. ☐

Exercise 21.25. Does density mean equality?

Hint. No. ☐

Solution. Arbitrarily close approximation. ☐

Exercise 21.26. Does uniform approximation imply integral approximation?

Hint. Yes. ☐

Solution. Yes on compact intervals. ☐

Exercise 21.27. Can $|x|$ be polynomially approximated uniformly?

Hint. Yes. ☐

Solution. Yes. ☐

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 22

Chebyshev and Minimax Approximation

Chebyshev and minimax approximation focus on worst-case uniform error. Equioscillation and Chebyshev polynomials reveal why optimal approximants spread their error rather than concentrating it.

Learning goals

The reader should be able to use the central definitions, prove the structural results, diagnose the role of each hypothesis, and solve representative analytic problems.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

22.1 Minimax problem

Choose an approximant minimizing the maximum absolute error.

22.2 Chebyshev polynomials

$T_n(\cos\theta) = \cos(n\theta)$ oscillates between ± 1 with controlled leading coefficient.

22.3 Extremal property

Among monic degree-n polynomials, a scaled Chebyshev polynomial minimizes sup norm on $[-1, 1]$.

22.4 Equioscillation

Optimal errors alternate in sign at enough extremal points.

22.5 Uniqueness

Alternation forces uniqueness of the best approximant in classical polynomial spaces.

22.6 Node placement

Chebyshev extrema and zeros guide stable interpolation and quadrature.

22.7 Error scaling

Affine changes transfer results from $[-1, 1]$ to any compact interval.

22.8 Numerical viewpoint

Minimax approximation optimizes worst-case performance rather than average error.

22.9 Core structural results

Theorem 22.1 (Chebyshev extremal property)**.** *The monic polynomial $2^{1-n}T_n$ has minimal sup norm among all monic degree-n polynomials on $[-1, 1]$.*

Proof. If another monic polynomial had smaller norm, their difference would have degree at most $n-1$ yet would change sign between the $n+1$ alternating extrema of the Chebyshev polynomial, forcing at least n roots, contradiction. □

Theorem 22.2 (Alternation criterion)**.** *A best degree-at-most-n uniform approximant to a continuous function is characterized by at least $n+2$ alternating extremal error points.*

Proof. If alternation fails, one can perturb within the approximation space to reduce the maximum error; if alternation holds, any competitor difference has too many sign changes for its degree. □

Theorem 22.3 (Uniqueness of minimax polynomial)**.** *The best uniform polynomial approximant of prescribed degree is unique.*

Proof. Two distinct best approximants averaged together would force a smaller maximum error unless their error extrema align; alternation then forces their difference to have too many zeros. □

22.10 Worked examples

Example 22.4 (Compute low Chebyshev polynomials)**.** Find T_0, T_1, T_2, T_3. From $T_n(\cos\theta) = \cos n\theta$: $1, x, 2x^2 - 1, 4x^3 - 3x$.

Example 22.5 (Recurrence)**.** Derive $T_{n+1} = 2xT_n - T_{n-1}$. Use $\cos((n+1)\theta) = 2\cos\theta\cos n\theta - \cos((n-1)\theta)$.

Example 22.6 (Extrema)**.** Where does T_n attain ± 1? At $x_k = \cos(k\pi/n)$, $k = 0, \dots, n$.

Example 22.7 (Zeros)**.** Where are the zeros of T_n? At $\cos((2k-1)\pi/(2n))$, $k = 1, \dots, n$.

22.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 22.8 (Compute low Chebyshev polynomials). Find T_0, T_1, T_2, T_3.

Solution. From $T_n(\cos\theta) = \cos n\theta$: $1, x, 2x^2 - 1, 4x^3 - 3x$. □

Problem 22.9 (Recurrence). Derive $T_{n+1} = 2xT_n - T_{n-1}$.

Solution. Use $\cos((n+1)\theta) = 2\cos\theta\cos n\theta - \cos((n-1)\theta)$. □

Problem 22.10 (Extrema). Where does T_n attain ± 1?

Solution. At $x_k = \cos(k\pi/n)$, $k = 0, \ldots, n$. □

Problem 22.11 (Zeros). Where are the zeros of T_n?

Solution. At $\cos((2k-1)\pi/(2n))$, $k = 1, \ldots, n$. □

Problem 22.12 (Monic scaling). Why multiply by 2^{1-n}?

Solution. For $n \geq 1$, the leading coefficient of T_n is 2^{n-1}, so scaling makes it monic. □

Problem 22.13 (Worst-case interpretation). What does minimax optimize?

Solution. The largest absolute error over the entire interval. □

Problem 22.14 (Alternating errors). Why should an optimum touch both positive and negative error extremes?

Solution. If all extreme errors had the same sign, a small vertical shift would reduce the maximum magnitude. □

Problem 22.15 (Linear approximation intuition). For degree one, how many alternating extremal points characterize the minimax fit?

Solution. At least three. □

Problem 22.16 (Affine rescaling). How do Chebyshev points transfer to $[a, b]$?

Solution. Map $x \in [-1, 1]$ by $(a+b)/2 + (b-a)x/2$. □

Problem 22.17 (Interpolation connection). Why do Chebyshev nodes reduce endpoint problems?

Solution. They cluster near endpoints, reducing the maximum node-polynomial magnitude. □

Problem 22.18 (Uniqueness mechanism). How does root counting enter minimax uniqueness?

Solution. Alternating error inequalities force the difference of two candidates to change sign too many times for its degree. □

Problem 22.19 (Minimax synthesis). How is minimax different from least squares?

Solution. Minimax controls the maximum error; least squares minimizes an integrated squared error and may tolerate larger localized peaks. □

22.12 Exercises with complete solutions

Exercise 22.20. What is T_2?

Hint. Use recurrence. □

Solution. $2x^2 - 1$. □

Exercise 22.21. What is T_3?

Hint. Use recurrence. □

Solution. $4x^3 - 3x$. □

Exercise 22.22. What is $\|T_n\|_\infty$ on $[-1, 1]$?

Hint. Cosine definition. □

Solution. 1. □

Exercise 22.23. How many extrema are used in alternation for degree n?

Hint. One more than dimension. □

Solution. At least $n + 2$. □

Exercise 22.24. Are Chebyshev zeros equally spaced in x?

Hint. They are equally spaced in angle. □

Solution. No. □

Exercise 22.25. Why scale T_n to monic?

Hint. Compare monic polynomials. □

Solution. To formulate the extremal property. □

Exercise 22.26. What norm is minimized?

Hint. Sup norm. □

Solution. Uniform norm. □

Exercise 22.27. Does minimax equal least squares?

Hint. Generally no. □

Solution. No. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 23

The Alternation Principle

The alternation principle is the structural certificate for best uniform approximation. It converts an optimization problem into a finite pattern of equal-magnitude alternating errors.

Learning goals

The reader should be able to use the central definitions, prove the structural results, diagnose the role of each hypothesis, and solve representative analytic problems.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

23.1 Error function

For approximant p, study $e = f - p$.

23.2 Extreme points

Minimax optimality is witnessed where $|e|$ reaches its sup norm.

23.3 Alternation pattern

Signs alternate at successive extremal points.

23.4 Necessity

Without enough alternation one can construct a perturbation reducing the maximum error.

23.5 Sufficiency

Enough alternating extrema block every lower-degree perturbation by root counting.

23.6 Uniqueness

Two best approximants would force their difference to have too many zeros.

23.7 Remez idea

Algorithms update candidate alternation points until equal error levels stabilize.

23.8 Approximation certificates

Alternation gives a verifiable finite certificate of global optimality.

23.9 Core structural results

Theorem 23.1 (Chebyshev alternation theorem)**.** *A polynomial p of degree at most n is best uniform approximation to continuous f on $[a, b]$ iff the error attains its maximum magnitude at at least $n + 2$ ordered points with alternating signs.*

Proof. Sufficiency follows because any better competitor would force the difference polynomial to change sign between each adjacent pair, producing more roots than its degree. Necessity follows from a separation/perturbation argument when no such alternating set exists. ☐

Theorem 23.2 (Uniqueness)**.** *The best degree-n uniform polynomial approximant is unique.*

Proof. If two best approximants existed, alternation for one and the competitor inequalities would force their difference to have at least $n + 1$ zeros despite degree at most n. ☐

Theorem 23.3 (Finite certificate)**.** *Once $n + 2$ alternating equal-magnitude error points are verified, global minimax optimality follows.*

Proof. This is the sufficiency direction of the alternation theorem. ☐

23.10 Worked examples

Example 23.4 (Constant approximation)**.** How many alternating points certify a best constant approximant? Two points with errors $+E$ and $-E$; the best constant centers the range between maximum and minimum values.

Example 23.5 (Best constant formula)**.** For continuous f, find the best constant in sup norm. It is $(M + m)/2$, where $M = \max f$ and $m = \min f$, with error $(M - m)/2$.

Example 23.6 (Degree-one certificate)**.** How many alternating extrema certify a best line? Three.

Example 23.7 (Why equal magnitude?)**.** Why must active extrema in the alternation pattern have the same magnitude? If one were strictly smaller, the true sup norm would be determined elsewhere and the finite optimality certificate would not be balanced.

23.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 23.8 (Constant approximation)**.** How many alternating points certify a best constant approximant?

Solution. Two points with errors $+E$ and $-E$; the best constant centers the range between maximum and minimum values. ☐

Problem 23.9 (Best constant formula)**.** For continuous f, find the best constant in sup norm.

Solution. It is $(M+m)/2$, where $M = \max f$ and $m = \min f$, with error $(M-m)/2$. ☐

Problem 23.10 (Degree-one certificate)**.** How many alternating extrema certify a best line?

Solution. Three. ☐

Problem 23.11 (Why equal magnitude?)**.** Why must active extrema in the alternation pattern have the same magnitude?

Solution. If one were strictly smaller, the true sup norm would be determined elsewhere and the finite optimality certificate would not be balanced. ☐

Problem 23.12 (Root-count sufficiency)**.** Explain the sign-change argument.

Solution. A uniformly better competitor must lie closer to f at each alternating extremum, forcing the difference of the two approximants to alternate sign between consecutive extremal points and thus have too many roots. ☐

Problem 23.13 (Uniqueness)**.** Why can two different minimax polynomials not share the same alternation certificate?

Solution. Their difference would need to vanish or change sign in too many intervals for its degree. ☐

Problem 23.14 (Remez iteration)**.** What does a Remez step update?

Solution. It solves for an approximant and common error level on a chosen alternating set, then replaces points using extrema of the new error. ☐

Problem 23.15 (Error centering)**.** For constant approximation, why center the maximum and minimum?

Solution. Any shift away from the midpoint increases the larger of the two endpoint range deviations. ☐

Problem 23.16 (Monotone target)**.** If f is monotone, where are extreme errors for the best constant?

Solution. At points attaining the minimum and maximum of f. ☐

Problem 23.17 (Certificate versus search)**.** Why is alternation valuable computationally?

Solution. Searching for the optimum is global, but once a candidate displays the required alternation, optimality can be certified finitely. ☐

Problem 23.18 (Generalized spaces)**.** What property of polynomials enables alternation arguments?

Solution. A nonzero degree-n polynomial cannot have more than n roots; more generally Chebyshev systems have controlled zero counts. ☐

Problem 23.19 (Alternation synthesis)**.** What does the theorem reveal conceptually?

Solution. Best worst-case approximations distribute their unavoidable error evenly with alternating sign rather than allowing one-sided or localized excess. ☐

23.12 Exercises with complete solutions

Exercise 23.20. Best constant to f uses which values?

Hint. Maximum and minimum. □

Solution. Their midpoint. □

Exercise 23.21. How many points for degree n?

Hint. $n + 2$. □

Solution. At least $n + 2$. □

Exercise 23.22. What alternates?

Hint. Error sign. □

Solution. Signs of extremal error. □

Exercise 23.23. What is equal?

Hint. Error magnitude. □

Solution. The sup error magnitude. □

Exercise 23.24. What proves sufficiency?

Hint. Root count. □

Solution. Sign changes and root counting. □

Exercise 23.25. What algorithm uses alternation?

Hint. Remez. □

Solution. Remez algorithm. □

Exercise 23.26. Is minimax best approximant unique?

Hint. For polynomials, yes. □

Solution. Yes. □

Exercise 23.27. What norm is involved?

Hint. Sup norm. □

Solution. Uniform norm. □

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 24

Numerical Quadrature

Numerical quadrature replaces an integral by a weighted finite sum of function values. Interpolation gives a systematic derivation, while error formulas explain exactness and the role of node placement.

Learning goals

The reader should be able to use the central definitions, prove the structural results, diagnose the role of each hypothesis, and solve representative analytic problems.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

24.1 Quadrature rules

Approximate $\int_a^b f$ by $\sum w_i f(x_i)$.

24.2 Interpolatory quadrature

Integrate an interpolation polynomial exactly to obtain weights.

24.3 Newton–Cotes rules

Equally spaced nodes yield midpoint, trapezoidal, Simpson, and higher rules.

24.4 Degree of exactness

A rule has degree m if it integrates all polynomials through degree m exactly.

24.5 Error formulas

Remainders depend on higher derivatives and powers of step size.

24.6 Composite rules

Apply a low-order rule repeatedly on a fine partition.

24.7 Gaussian quadrature

Optimally chosen nodes increase polynomial exactness for a fixed number of samples.

24.8 Stability and adaptivity

Practical quadrature balances truncation error, smoothness, and sampling cost.

24.9 Core structural results

Theorem 24.1 (Trapezoidal error). *For $f \in C^2[a,b]$, the one-panel trapezoidal error equals $-(b-a)^3 f''(\xi)/12$ for some ξ.*

Proof. Integrate the linear interpolation remainder or use a Peano-kernel argument. □

Theorem 24.2 (Simpson exactness). *Simpson's rule is exact for polynomials of degree at most three.*

Proof. Check the rule on the basis $1, x, x^2, x^3$ after affine reduction to a symmetric interval. □

Theorem 24.3 (Gaussian exactness). *An n-node Gaussian rule for a positive weight is exact for polynomials through degree $2n-1$.*

Proof. Orthogonal-polynomial nodes make the error polynomial divisible by the degree-n orthogonal polynomial; division separates any degree-$2n-1$ polynomial into a vanishing part plus a lower-degree remainder integrated exactly by the weights. □

24.10 Worked examples

Example 24.4 (Midpoint rule). Approximate $\int_0^1 x^2 dx$ by one midpoint sample. The midpoint value is $1/4$, so the one-panel approximation is $1/4$ versus exact $1/3$.

Example 24.5 (Trapezoidal rule). Approximate the same integral. $(f(0)+f(1))/2 = 1/2$.

Example 24.6 (Simpson rule). Apply Simpson to x^2 on $[0,1]$. $(1/6)(f(0)+4f(1/2)+f(1)) = (1/6)(0+1+1) = 1/3$, exact.

Example 24.7 (Degree of exactness). Why is trapezoidal exact for linear functions? It integrates the linear interpolant, which equals the function itself for degree at most one.

24.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 24.8 (Midpoint rule). Approximate $\int_0^1 x^2 dx$ by one midpoint sample.

Solution. The midpoint value is $1/4$, so the one-panel approximation is $1/4$ versus exact $1/3$. □

Problem 24.9 (Trapezoidal rule)**.** Approximate the same integral.

Solution. $(f(0) + f(1))/2 = 1/2$. ☐

Problem 24.10 (Simpson rule)**.** Apply Simpson to x^2 on $[0, 1]$.

Solution. $(1/6)(f(0) + 4f(1/2) + f(1)) = (1/6)(0 + 1 + 1) = 1/3$, exact. ☐

Problem 24.11 (Degree of exactness)**.** Why is trapezoidal exact for linear functions?

Solution. It integrates the linear interpolant, which equals the function itself for degree at most one. ☐

Problem 24.12 (Composite trapezoid)**.** What changes when the interval is subdivided?

Solution. Apply trapezoidal rule on each subinterval and sum; the global error decreases like h^2 under smoothness. ☐

Problem 24.13 (Composite Simpson)**.** What order is expected?

Solution. For sufficiently smooth functions the global error behaves like h^4. ☐

Problem 24.14 (Interpolatory weights)**.** How are weights obtained?

Solution. Integrate each Lagrange cardinal polynomial over the interval. ☐

Problem 24.15 (Positive weights)**.** Why are positive quadrature weights desirable?

Solution. They improve stability and preserve positivity for nonnegative sampled data. ☐

Problem 24.16 (Gaussian advantage)**.** Why can n Gaussian nodes beat n equally spaced interpolatory nodes?

Solution. The nodes are optimized jointly with weights, yielding degree of exactness $2n-1$. ☐

Problem 24.17 (Error and smoothness)**.** Why do high-order rules need higher derivatives?

Solution. Their error formulas involve higher derivatives; without regularity, formal high order may not translate into actual accuracy. ☐

Problem 24.18 (Adaptive idea)**.** How does adaptive quadrature choose where to sample more?

Solution. Estimate local error on subintervals and refine where the estimate is largest. ☐

Problem 24.19 (Quadrature synthesis)**.** How is quadrature connected to interpolation?

Solution. Interpolatory quadrature integrates a polynomial surrogate exactly; the quadrature error is therefore the integral of the interpolation error. ☐

24.12 Exercises with complete solutions

Exercise 24.20. Degree of trapezoidal exactness?

Hint. Linear. ☐

Solution. 1. ☐

Exercise 24.21. Degree of Simpson exactness?

Hint. Cubic. ☐

Solution. 3. ☐

Exercise 24.22. What does Gaussian n-point exactness reach?

Hint. $2n - 1$. ☐

Solution. Degree $2n - 1$. ☐

Exercise 24.23. Why composite rules?

Hint. Reduce step size. ☐

Solution. To lower error on long intervals. ☐

Exercise 24.24. What are quadrature weights?

Hint. Integrated basis functions. ☐

Solution. Coefficients multiplying samples. ☐

Exercise 24.25. Is Simpson exact for x^4?

Hint. No. ☐

Solution. Not generally. ☐

Exercise 24.26. What smoothness controls trapezoidal error?

Hint. Second derivative. ☐

Solution. A bound on f''. ☐

Exercise 24.27. What is adaptive refinement based on?

Hint. Local error estimates. ☐

Solution. Estimated local error. ☐

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.

Chapter 25

Continued Fractions and Approximation Topics

Continued fractions provide a discrete approximation theory for real numbers. Their convergents are exceptionally good rational approximants and connect Euclidean algorithms, Diophantine approximation, and recursive computation.

Learning goals

The reader should be able to use the central definitions, prove the structural results, diagnose the role of each hypothesis, and solve representative analytic problems.

Conceptual roadmap

$$\boxed{\text{definition} \longrightarrow \text{estimate} \longrightarrow \text{theorem} \longrightarrow \text{application.}}$$

25.1 Simple continued fractions

Write a real number as $[a_0; a_1, a_2, \ldots]$ with positive integer partial quotients after the first.

25.2 Euclidean algorithm

Rational continued fractions terminate and encode the Euclidean algorithm.

25.3 Convergents

Truncating the expansion gives rational approximants p_n/q_n.

25.4 Recurrences

Numerators and denominators satisfy second-order recurrences.

25.5 Determinant identity

Adjacent convergents satisfy $p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$.

25.6 Approximation quality

Convergents approximate unusually well relative to their denominators.

25.7 Quadratic irrationals

Eventually periodic continued fractions characterize quadratic irrationals.

25.8 Approximation perspective

Continued fractions complement polynomial approximation by optimizing arithmetic rather than functional error.

25.9 Core structural results

Theorem 25.1 (Convergent recurrences)**.** *With standard initial data,* $p_n = a_n p_{n-1} + p_{n-2}$ *and* $q_n = a_n q_{n-1} + q_{n-2}$.

Proof. Induct by rewriting the finite continued fraction after adjoining its last partial quotient. □

Theorem 25.2 (Adjacent determinant identity)**.** $p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$.

Proof. Insert the recurrences and observe the determinant changes sign at each step. □

Theorem 25.3 (Basic approximation bound)**.** *For an irrational simple continued fraction,* $|x - p_n/q_n| < 1/q_n^2$ *for infinitely many convergents, and standard sharper neighboring-denominator bounds hold for every convergent.*

Proof. Express x using the complete quotient after stage n and combine with the determinant identity; the denominator in the exact error formula exceeds q_n^2 at the appropriate stages. □

25.10 Worked examples

Example 25.4 (Rational expansion)**.** Find a continued fraction for 17/7. Euclidean algorithm: $17 = 2 \cdot 7 + 3$, $7 = 2 \cdot 3 + 1$, $3 = 3 \cdot 1$, so $17/7 = [2; 2, 3]$.

Example 25.5 (Convergents of $[1; 1, 1, \ldots]$)**.** Identify the convergents. They are ratios of consecutive Fibonacci numbers.

Example 25.6 (Golden ratio)**.** Show $\varphi = [1; 1, 1, 1, \ldots]$. The tail equals the whole number, so $x = 1 + 1/x$, giving $x^2 - x - 1 = 0$ and the positive root φ.

Example 25.7 (Recurrence computation)**.** Given partial quotients a_0, a_1, a_2, compute p_2, q_2 recursively. Use $p_{-2} = 0, p_{-1} = 1$, $q_{-2} = 1, q_{-1} = 0$, then apply the stated recurrences.

25.11 Solved dossiers

The dossiers are longer solved problems. Legacy mappings guide topic selection where explicit retained source blocks exist; otherwise fresh canonical problems complete the chapter.

Problem 25.8 (Rational expansion)**.** Find a continued fraction for 17/7.

Solution. Euclidean algorithm: $17 = 2 \cdot 7 + 3$, $7 = 2 \cdot 3 + 1$, $3 = 3 \cdot 1$, so $17/7 = [2; 2, 3]$. □

Problem 25.9 (Convergents of $[1; 1, 1, \dots]$)**.** Identify the convergents.

Solution. They are ratios of consecutive Fibonacci numbers. □

Problem 25.10 (Golden ratio)**.** Show $\varphi = [1; 1, 1, 1, \dots]$.

Solution. The tail equals the whole number, so $x = 1 + 1/x$, giving $x^2 - x - 1 = 0$ and the positive root φ. □

Problem 25.11 (Recurrence computation)**.** Given partial quotients a_0, a_1, a_2, compute p_2, q_2 recursively.

Solution. Use $p_{-2} = 0, p_{-1} = 1$, $q_{-2} = 1, q_{-1} = 0$, then apply the stated recurrences. □

Problem 25.12 (Neighbor difference)**.** Use the determinant identity to compute $|p_n/q_n - p_{n-1}/q_{n-1}|$.

Solution. It equals $1/(q_n q_{n-1})$. □

Problem 25.13 (Why denominators grow)**.** Show $q_n > q_{n-1}$ for simple continued fractions beyond the first stage.

Solution. Since $a_n \geq 1$ and $q_{n-2} \geq 0$, $q_n = a_n q_{n-1} + q_{n-2} \geq q_{n-1}$, with strict growth once positive preceding terms occur. □

Problem 25.14 (Best-approximation intuition)**.** Why are convergents hard to beat?

Solution. The determinant identity forces nearby rational competitors with small denominators to remain separated, while the continued-fraction error is already on the order of inverse denominator squared. □

Problem 25.15 (Finite expansion ambiguity)**.** Why can a rational have two finite simple continued fraction expansions?

Solution. A final partial quotient $a_n > 1$ can be replaced by $a_n - 1, 1$, producing the same value. □

Problem 25.16 (Square root periodicity)**.** What is special about continued fractions of quadratic irrationals?

Solution. Their partial quotients are eventually periodic, and conversely eventual periodicity characterizes quadratic irrationals. □

Problem 25.17 (Euclidean algorithm link)**.** Why do rationals terminate?

Solution. Repeated reciprocal-and-floor steps reproduce Euclidean remainders, which eventually reach zero. □

Problem 25.18 (Approximation scale)**.** What does an error of order $1/q^2$ mean?

Solution. Doubling denominator can reduce error quadratically in the ideal scale, much better than a generic grid spacing of order $1/q$. □

Problem 25.19 (Continued-fraction synthesis)**.** How does this chapter fit the approximation part of the volume?

Solution. Earlier chapters approximate functions globally; continued fractions approximate numbers arithmetically, using recursive discrete structure to obtain near-optimal rational approximants. □

25.12 Exercises with complete solutions

Exercise 25.20. Find CF of 3/2.

Hint. Euclidean algorithm. ☐

Solution. $[1; 2]$. ☐

Exercise 25.21. What are convergents?

Hint. Finite truncations. ☐

Solution. Rational truncations of the continued fraction. ☐

Exercise 25.22. What algorithm computes rational CFs?

Hint. Euclidean algorithm. ☐

Solution. Euclidean algorithm. ☐

Exercise 25.23. What famous CF has all ones?

Hint. Golden ratio. ☐

Solution. φ. ☐

Exercise 25.24. How do q_n behave?

Hint. They grow. ☐

Solution. They increase after the initial stages. ☐

Exercise 25.25. What identity relates adjacent convergents?

Hint. Determinant identity. ☐

Solution. $p_n q_{n-1} - p_{n-1} q_n = \pm 1$. ☐

Exercise 25.26. What numbers have eventually periodic CFs?

Hint. Quadratic irrationals. ☐

Solution. Exactly quadratic irrationals. ☐

Exercise 25.27. What error scale do convergents achieve?

Hint. About inverse-square denominator. ☐

Solution. Order $1/q_n^2$. ☐

Chapter summary

The chapter's tools now form part of the canonical Volume II analysis toolkit and will be used in later approximation, fixed-point, and convergence arguments.